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Arbitrage-free deep option pricing via convex composition: architectural guarantees and universal approximation

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ABSTRACT
We introduce ArbNet, a neural call-pricing architecture that is free of static arbitrage *by construction* rather than by penalisation. The European call price is parameterised as a forward-intrinsic baseline plus a non-negative correction built from a mixture of products of input-convex networks in strike and monotone networks in time-to-maturity, modulated by a boundary factor that vanishes at expiry. Elementary convexity and monotonicity compositions deliver four pointwise no-arbitrage conditions simultaneously — butterfly, calendar, expiry boundary, and forward-intrinsic lower bound — for every choice of weights, with no tolerance-based residual. We prove this guarantee together with a universal-approximation theorem on the cone of continuous, strike-convex, time-monotone, non-negative surfaces vanishing at expiry, and an explicit negative companion: surfaces whose forward time value is non-convex in strike, the entire Black–Scholes family among them, lie outside the hypothesis class. In a 1,359-day walk-forward over NSE Nifty 50 index options (2019–2024), ArbNet records zero butterfly and zero calendar violations on *every* day; a faithful soft-penalty baseline violates static arbitrage on 961 days. The guarantee costs about 59 INR per call in mean RMSE (Diebold–Mariano 32.1, $p < 10^{-3}$) — a structural trade-off, suited to consumers that require certified no-arbitrage at every grid evaluation, independently of the training trajectory.

KEYWORDS
No-arbitrage; neural option pricing; input-convex networks; monotone networks; universal approximation; risk-neutral density

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C45; G13

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1. Introduction

A European call surface — the map $(K, T) \mapsto C(K, T)$ on strike and maturity at a fixed valuation date — is the central object on which a derivative desk does its work. It feeds three distinct downstream tasks: hedging, through the spot, strike, and second-strike derivatives that yield delta and gamma; book-marking and capital reporting, through the implied risk-neutral density extracted by Breeden–Litzenberger (Breeden and Litzenberger, 1978) and consumed by value-at-risk and stress-test pipelines; and exotic-product calibration, since variance swaps, barriers, and cliquets all require a consistent vanilla surface to extract a unique implied density. Each of these consumers depends on the surface being *free of static arbitrage*: no portfolio of vanilla calls assembled at the valuation date should produce a non-negative future payoff at strictly negative cost today.

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Static no-arbitrage decomposes into four pointwise shape constraints on $C(\cdot,\cdot)$ (Cousot, 2007; Roper, 2010; Davis and Hobson, 2007). Convexity in strike at every maturity is the analytic dual of the butterfly portfolio, and is the failure mode most often documented on stressed surfaces (Kahalé, 2004; Fengler, 2009). Carry-adjusted calendar monotonicity — $e^{rT}C(K,T)$ non-decreasing in T at fixed K — is dual to the calendar spread, and its violation arbitrages options of different maturities against each other. The forward-intrinsic lower bound $C(K,T) \geq e^{-rT}(F_T - K)^+$, with $F_T = Se^{(r-q)T}$, is the static replication of the forward via put-call parity (Merton, 1973). The fourth, often left implicit, is the expiry boundary $C(K,0) = (S-K)^+$. Together they are the canonical structural list against which any vanilla surface is judged, and the list Section 3 states formally.

Classical financial theory provides parametric models that are arbitrage-free by construction. The Black–Scholes model (Black and Scholes, 1973) is arbitrage-free at any single volatility but cannot fit empirical surfaces, which exhibit pronounced smiles and skews (Cont, 2001; Rebonato, 2004). The local-volatility model of Dupire (Dupire, 1994) fits perfectly by construction but has been shown to mismatch dynamic properties of the surface (Cont and da Fonseca, 2002; Hagan et al., 2002). Stochastic-volatility models (Heston (Heston, 1993), SABR (Hagan et al., 2002), and their many extensions (Andersen and Piterbarg, 2007)) are dynamically arbitrage-free and have analytically tractable transforms (Lewis, 2001; Carr and Madan, 1999; Eberlein et al., 2010), but their fit to empirical surfaces requires either many free parameters (Andreasen and Huge, 2011) or model extensions such as jumps (Carr et al., 2002; Cont and Tankov, 2003). Rough-volatility models (Alòs et al., 2007; Bayer et al., 2016, 2023; Bennedsen et al., 2017; El Euch and Rosenbaum, 2018) achieve realistic ATM volatility behaviour with few parameters but have non-trivial calibration costs (Horvath et al., 2021). At the surface level, the SVI (Gatheral, 2004) and SSVI (Gatheral and Jacquier, 2014a) parameterisations of total implied variance are arbitrage-free under explicit parameter constraints (Gatheral and Jacquier, 2014b; Gatheral et al., 2018); they are widely deployed in practice but have a finite-dimensional capacity that constrains the fit on stressed wings.

The advent of deep learning has produced an array of neural pricers (Ackerer et al., 2020; Chataigner et al., 2020; Horvath et al., 2021; Ruf and Wang, 2020b,a; Hutchinson et al., 1994; Itkin, 2019; Sirignano, 2019; Sirignano and Cont, 2019; Liu et al., 2019; Ferguson and Green, 2018), which trade the parametric tractability of classical models for the capacity of multi-layer architectures. These pricers achieve excellent in-sample fit and, with enough training, decent out-of-sample generalisation. However, almost all neural pricers enforce no-arbitrage *softly*: an additional term proportional to the negative part of the local arbitrage violation is added to the training loss, typically as $\lambda_{\mathrm{bf}}\,\mathbb{E}[(-\partial_K^2 C_\theta)_+^2]$ for the butterfly condition and $\lambda_{\mathrm{cal}}\,\mathbb{E}[(\partial_T(e^{rT}C_\theta))_-^2]$ for the calendar condition, with the expectations approximated by Monte Carlo on a finite training grid. The soft penalty paradigm has a number of well-understood limits.

First, the penalty is a regulariser, not a constraint. At any finite penalty weight λ, the optimum trades off the pricing-error term against the penalty term, and as long as the pricing-error gradient pulls the surface into the arbitrage region anywhere, some violation will persist. In our experiments, with the standard penalty weight $\lambda = 1$ and training budgets in the hundreds of epochs, the soft-penalty baseline still violates static arbitrage on 961 of the 1,359 real trading days of our walk-forward study (Section 8.1).

Second, the penalty is evaluated on the training grid, not over the continuous domain. The neural pricer is queried at many points off-grid by downstream consumers — e.g., at the strikes of exotic-product instruments, at fine grids for risk-neutral density extraction, or at perturbations of the spot for delta hedging — and at those off-grid points, no penalty was applied during training. A surface that is compliant on the training grid can therefore violate the conditions arbitrarily far off-grid. This is the noisy-empirical-grid analogue of the more general phenomenon studied by Aït-Sahalia and Duarte (2003) in the context of shape-constrained nonparametric pricing.

Third, residual violations propagate. A non-convexity in K at one grid point implies a negative second strike derivative, which in turn implies a negative implied density at that point. The negative density propagates into any computation that uses the surface — expected payoffs of exotic products, value-at-risk computations, model-free volatility indices — with potentially significant aggregate effects. Hedging with a delta extracted from a non-convex surface introduces additional path-dependent variance because the delta is no longer monotone in strike.

The natural alternative is to make the architecture itself arbitrage-free. This idea has historical precedent. SVI (Gatheral, 2004) and SSVI (Gatheral and Jacquier, 2014a) are structurally arbitrage-free under parameter constraints in the implied-variance domain. In the call-price domain, Kahalé (2004) and Fengler (2009) construct arbitrage-free interpolators of finite quote sets via spline-based monotone-convex schemes; the constructions are exact but not learnable from data at the level of a neural model. Density-network approaches (Horvath et al., 2021) parameterise the risk-neutral density ρ_T directly; this trivially enforces butterfly but does not give a closed-form calendar enforcement. Neural stochastic differential equation (neural-SDE) market models (Cohen et al., 2023, 2022; Cuchiero et al., 2020; Gambara and Teichmann, 2021) build arbitrage-free dynamics on the underlying process; static no-arbitrage is then a corollary. This last line is the most theoretically satisfying but is also the most computationally costly and the least transparent to inspect at the level of a single surface.

The structural literature reviewed above lacks a single construction that combines, in one design, the four properties that a deployable neural pricer in the call-price domain needs. *Structural compliance* requires that the four pointwise no-arbitrage conditions hold by the parameterisation itself, at every choice of weights, rather than be approximated through a penalty term in the loss. *Neural capacity* requires that the construction carry enough degrees of freedom to fit stressed real surfaces in the wings, rather than the handful of coefficients of a parametric form. *Single-pass pricing* requires that a price be recovered by one forward evaluation of the network, with Greeks by vectorised autodifferentiation, free of path simulation or numerical integration at query time. And a *delimiting representational guarantee* requires an explicit statement of both what the construction can and what it cannot reproduce. Each existing approach acquires a strict subset of these four. SVI and SSVI carry structural compliance, single-pass pricing, and a finite-capacity analogue of the guarantee, but cannot scale capacity. The arbitrage-free interpolators of Kahalé (2004) and Fengler (2009) carry compliance and single-pass pricing on the quoted set, but provide no learnable surface beyond it. Density-network pricers carry capacity and the butterfly part of compliance, but enforce calendar monotonicity through a penalty and recover a call price only by numerical integration. Neural-SDE market models carry compliance in its strongest form — static *and* dynamic — together with capacity, but pay for it with per-query path simulation and offer no transparent surface-level representational result. The empty corner, where all four properties hold jointly and the empirical claim is checked on real, noisy cross-sections rather than synthetic surfaces alone, is the gap that this paper closes.

We make two contributions, each supported by a theorem with proof, and together they close the four properties of the previous paragraph.

Contribution 1 (closing structural compliance, neural capacity, and single-pass pricing). The first contribution is the architecture itself, ArbNet. We parameterise the European call price as

$$C_\theta(K,T) \;=\; e^{-rT}\big[\,(F_T-K)^+ + \Delta_\theta(K,T)\big]\,, \qquad F_T = S\,e^{(r-q)T}, \tag{1}$$

with a non-negative correction Δ_θ given by

$$\Delta_\theta(K,T) = S\,(1-e^{-\alpha_\theta T})\sum_{j=1}^{J} s_{j,\theta}\ \mathrm{softplus}\big(\phi_j^{\mathrm{cvx}}(K/S)\big)\ \mathrm{softplus}\big(\phi_j^{\mathrm{mon}}(T)\big), \tag{2}$$

where each ϕ_j^{cvx} is an input-convex network (Amos et al., 2017; Bunne et al., 2022; Chen et al., 2019) in normalised strike, each ϕ_j^{mon} is a monotone network (Daniels and Velikova, 2010; Sill, 1998; Wehenkel and Louppe, 2019; Runje and Shankaranarayana, 2023; Guo and Lan, 2020) in time-to-maturity, and the scalar gates satisfy $s_{j,\theta}, \alpha_\theta > 0$. Theorem 5.4 establishes that the resulting surface satisfies the four pointwise no-arbitrage conditions for every $\theta \in \Theta$, with no tolerance bound; this is structural compliance imposed by the parameterisation itself, not by a regulariser in the loss. The default configuration carries on the order of 30,000 parameters, supplying the capacity that finite-dimensional structural models cannot. The forward pass is a single network evaluation, and the Greeks are recovered by vectorised autodifferentiation (Baydin et al., 2018), free of path simulation and numerical integration. The decision to work in the call-price domain rather than the implied-vol domain follows directly from the requirement of compliance under elementary compositions: as Section 4.5 explains, the implied-vol domain demands the non-linear Durrleman positivity condition, which is not preserved by the convex–monotone primitives we use.

Contribution 2 (closing the delimiting representational guarantee). The second contribution characterises the architecture's expressivity in both directions. The positive half is Theorem 5.10: the hypothesis class $\mathcal{C}_{\mathrm{cvx},\nearrow}$ of continuous, strike-convex, time-non-decreasing, non-negative surfaces vanishing at $T = 0$ on the compact box $\mathcal{B} = [K_0, K_1] \times [0, T_{\max}]$ is uniformly approximable by the parameterisation Equation (2), with mixture count scaling as $J = \mathcal{O}(\omega_g^{-1}(\varepsilon/3)^{-2})$ in the bivariate modulus of continuity ω_g of the target. The negative half is Theorem 5.11: surfaces whose forward time value is non-convex in strike, including the entire Black–Scholes family at any positive volatility, lie outside $\mathcal{C}_{\mathrm{cvx},\nearrow}$ and cannot be represented exactly. The two together discharge the fourth desideratum: a representational result that says, in one breath, both what the architecture can and what it cannot reproduce. The proof of universality combines a tensor-product density argument with the input-convex-network density result of Amos and Xu (2017) and the monotone-network density result of Daniels and Velikova (2010), in the universal-approximation lineage inaugurated by Cybenko (1989), Hornik (1989; 1991), Leshno et al. (1993), Pinkus (1999), and Yarotsky (2017).

To support these two contributions empirically, we run a controlled benchmark over 30 rough-Bergomi surfaces and a 1,359-day walk-forward over National Stock Exchange of India (NSE) Nifty 50 index options spanning 1 January 2019 to 5 July 2024. The architecture records identically zero butterfly and zero calendar violations on every trial and every day, while a faithful soft-penalty Ackerer-style baseline violates static arbitrage on 961 of those 1,359 days. The compliance guarantee carries a measurable price-fitting cost — on the real data ArbNet's mean daily price RMSE exceeds the soft-penalty baseline's by roughly 59 INR per call, overwhelmingly significant under a HAC-corrected Diebold–Mariano test (DM = 32.05, $p < 10^{-3}$) — and the trade-off is localised cleanly on a fit–compliance frontier (Figure 4). A self-contained PyTorch (Paszke et al., 2019) implementation reproduces every number reported here from the committed data and pinned random seeds, with the synthetic study running in under two minutes on a single CPU and the full walk-forward in a few CPU-hours.

To circumscribe our claims sharply, we are explicit about what we do *not* demonstrate. First, we do not claim uniform RMSE dominance over the soft-penalty baseline: as Table 3 shows, the soft-penalty baseline fits prices better on average, and a practitioner whose loss function is exclusively RMSE will prefer it. Second, we do not claim that the architecture can represent every plausibly-encountered call surface: surfaces whose forward time-value is non-convex in K, including the Black–Scholes family at any positive volatility (proved in Theorem 5.11), are outside the hypothesis class. Third, we treat only *static* no-arbitrage (i.e., the pointwise structural conditions on $C(\cdot, \cdot)$ at a single valuation date), not dynamic no-arbitrage (the requirement that there exist a consistent risk-neutral process driving the surface forward in time); the latter is the province of neural-SDE market models (Cohen et al., 2023; Cuchiero et al., 2020; Gambara and Teichmann, 2021). Fourth, our real-data study covers a single liquid index — NSE Nifty 50 — over one five-and-a-half-year window; we do not claim that the *magnitudes* (the size of the RMSE gap, the violation frequency of the soft-penalty baseline) transfer unchanged to other underlyings or periods. The qualitative conclusion — a hard guarantee for ArbNet against residual violations for soft penalties — is a property of the architecture and should transfer, but the numbers are market-specific.

The rest of the paper proceeds as follows. Section 2 reviews the literature on parametric, soft-penalty, and structurally-constrained pricers, drawing the conceptual map within which our contribution sits. Section 3 introduces notation and recalls the static no-arbitrage conditions with their economic interpretation; it also discusses the Lee moment formula (Lee, 2004, 2005) that constrains the wings of an arbitrage-free surface. Section 4 introduces the construction itself, with the computational graph (Figure 1), forward-pass pseudocode (Algorithm 1), motivating discussion for each design choice, and complexity analysis. Section 5 states and proves the two main theorems. Section 6 describes the training objective, the diagnostics, and the optimisation algorithm. Section 7 details the experimental protocol for both the synthetic and the real-data studies. Section 8 reports the findings with five figures and two result tables. Section 9 discusses limitations, comparisons to alternatives, and future directions. Section 10 concludes. Appendices contain detailed proofs, Greek calculations, the Newton inversion procedure, and the full hyper-parameter table.

2. Related work

The literature on neural option pricing organises naturally along two axes: *structural* versus *soft* arbitrage enforcement, and the *implied-vol* versus *call-price* domain. The classical parametric models occupy one corner of this map (structural compliance, parametric capacity), modern soft-penalty neural pricers another (large capacity, tolerance-based compliance), and a smaller but growing structurally-constrained literature occupies the same corner as our work. We review each in turn, with particular attention to where the domain choice forces or relieves the no-arbitrage enforcement.

2.1. *Parametric arbitrage-free models*

The classical lineage of arbitrage-free pricing begins with Black–Scholes (Black and Scholes, 1973; Merton, 1973), in which the assumption of geometric Brownian motion of the underlying yields a closed-form call price under a single volatility parameter σ. The model is structurally arbitrage-free and provides the language in which all subsequent surfaces are described: the call price determines an implied volatility $\sigma(K,T)$ via inversion of the Black–Scholes formula, and a static-arbitrage-free surface has a non-decreasing total implied variance $w(k,T) = \sigma^2 T$ in T and a Durrleman-positive shape in k (Gatheral and Jacquier, 2014b,a). The empirical inadequacy of Black–Scholes on real surfaces — the smile and skew, well documented since (Rebonato, 2004) — gave rise to three modelling traditions, each preserving structural arbitrage-freeness in its own way.

Local-volatility models (Dupire, 1994) parameterise the diffusion coefficient as a deterministic function of spot and time; the resulting call surface fits the observed surface exactly by construction. However, the dynamics of the local-volatility model are well-known to mismatch the empirical dynamics of the smile (Cont and da Fonseca, 2002); in particular, the smile dynamics under a local-volatility model are essentially backwards from observed market dynamics, a phenomenon documented by Hagan et al. (2002). Stochastic-volatility models such as Heston (Heston, 1993) and SABR (Hagan et al., 2002), and their many extensions (Andersen and Piterbarg, 2007; Andersen and Andreasen, 2000; Andreasen and Huge, 2011), restore plausible dynamics at the cost of giving up exact static fit; calibration trades off in-sample fit against parsimony. These models are dynamically arbitrage-free under explicit conditions on their parameters; Lewis (2001) and Carr–Madan (1999) gave efficient Fourier methods for their evaluation, and the analytical structure has been extended to Lévy and exponential-Lévy models (Carr et al., 2002; Cont and Tankov, 2003; Eberlein et al., 2010). Rough-volatility models (Alòs et al., 2007; Bayer et al., 2016; Bennedsen et al., 2017; Bayer et al., 2023; El Euch and Rosenbaum, 2018; Jacquier et al., 2018) take the Hurst parameter $H < 1/2$ as a structural feature and achieve realistic ATM-volatility behaviour, but calibration remains computationally non-trivial and has been a recent target of neural acceleration (Horvath et al., 2021; Bayer and Stemper, 2019).

At the surface level, the SVI parameterisation (Gatheral, 2004) models total implied variance per maturity as a five-parameter raw form, and SSVI (Gatheral and Jacquier, 2014a) extends this to a surface-wide three-parameter form. Both are arbitrage-free under explicit (and often non-tight) parameter constraints; Jacquier and Roome (2014b) catalogue these constraints, and Gatheral et al. (2018) extends the analysis. SVI and SSVI are widely deployed in practice. They sit conceptually closest to our construction in that they offer structural arbitrage-freeness via parameter constraints rather than via tolerance-based penalties; the difference is that their capacity is finite-dimensional (five or three parameters per surface), whereas our architecture has neural-network capacity.

2.2. *Soft-penalty neural pricers*

The pioneering work of Hutchinson, Lo, and Poggio (1994) applied a neural network to non-parametric option pricing, decades before the deep learning revolution. The modern wave begins with Ackerer, Tagasovska, and Vatter (2020), who train an unconstrained multi-layer perceptron $w_\theta(k, T)$ in log-moneyness coordinates as a model for the total implied variance, and add to the price-RMSE training loss two penalty terms,

$$\mathcal{L}_{\mathrm{soft}}(\theta) = \mathcal{L}_{\mathrm{price}}(\theta) + \lambda_{\mathrm{bf}} \sum_{(k,T)\in\mathcal{G}} (-\mathcal{D}(w_\theta)(k,T))_+ + \lambda_{\mathrm{cal}} \sum_{(k,T)\in\mathcal{G}} (-\partial_T w_\theta(k,T))_+, \tag{3}$$

where $\mathcal{D}$ is the Durrleman positivity operator and $\mathcal{G}$ is a training grid. Chataigner, Crépey, and Dixon (2020) extend the paradigm directly to the call-price domain. Horváth, Muguruza, and Tomas (2021) apply a similar architecture to deep pricing under rough-volatility models. Itkin (2019) surveys variants. The Ackerer-style soft-penalty baseline used in our experiments instantiates this paradigm faithfully, sharing the optimiser, learning-rate schedule, and training data of ArbNet. Ruf and Wang (2020b; 2020a) review the broader literature systematically.

All of these models share a structural limitation: the penalty is a regulariser, not a constraint. At any finite penalty weight and any finite training budget, the optimum is a stationary point of the regularised loss, which generically carries residual violations. The limit $\lambda \to \infty$ does not formally resolve this either, since the resulting constrained optimisation problem must be solved exactly — intractable for neural pricers, since the constraint must hold over a continuous rather than a finite-grid domain. The shape-constrained nonparametric programme of Aït-Sahalia and Duarte (2003) attempts an exact projection onto the convex cone of admissible surfaces, but at scales well below those required for production neural pricers.

2.3. *Structurally-constrained neural pricers*

A growing literature aims to build no-arbitrage into the architecture itself rather than into the loss, along three distinct lines. The first uses input-convex networks (ICNNs) (Amos et al., 2017) and monotone networks (Daniels and Velikova, 2010; Sill, 1998) as building blocks within the implied-vol domain. The obstacle is that the no-arbitrage condition there is the Durrleman positivity, a non-linear inequality coupling w, $\partial_k w$, and $\partial_k^2 w$ (Gatheral and Jacquier, 2014a), which is not preserved by simple convex–monotone composition. One must therefore either restrict to surfaces where Durrleman positivity is implied by more elementary parameter constraints (the SVI case) or accept residual violations. The variational-autoencoder approach of Bergeron et al. (2022) for IV surfaces and the generative models of Henry-Labordère (2019) for trading strategies sit at this interface.

The second approach builds neural stochastic differential equation (neural-SDE) market models on the underlying process (Cohen et al., 2023, 2022; Cuchiero et al., 2020; Gambara and Teichmann, 2021; Liu et al., 2019). Here the underlying process itself is a neural object, and arbitrage-freeness emerges as a consistency requirement on the drift and diffusion under the risk-neutral measure. The approach is theoretically clean and yields both static and dynamic compliance simultaneously, but it is computationally demanding (one must simulate paths under the learned SDE for every pricing query) and the inductive bias of the surface shape is harder to read off.

The third approach, including ours, parameterises the call price directly in the call-price domain and uses the elementary structural conditions of Theorem 3.1. This is feasible because, in the call-price domain, the no-arbitrage conditions are first-order linear (convexity in K, monotonicity in $e^{rT}C$ in T), which is exactly what ICNN and monotone network compositions preserve. Imaki et al. (2021) pursue a closely related agenda in the context of deep hedging; (Ferguson and Green, 2018) explore the connection between neural pricing and partial differential equation methods.

2.4. *Deep hedging*

Buehler, Gonon, Teichmann, and Wood (Buehler et al., 2019) introduced deep hedging, in which the hedging policy is learned directly without an intermediate pricing model. The pricing model is replaced by a market simulator (e.g., a calibrated rough-Bergomi process), and the policy network is trained to minimise a convex risk measure (CVaR, entropic risk) of the terminal hedged PnL. Subsequent work (Chakrabarti et al., 2019; Imaki et al., 2021; Vallarino, 2024) has refined this in directions including transaction costs and reinforcement-learning-style policy gradients. Deep hedging is conceptually orthogonal to our architecture: deep hedging gives a policy but no surface; ArbNet gives a surface (with structural compliance) from which a policy is extracted via autodifferentiation of $\partial_S C_\theta$ (Section B). The two approaches can therefore be combined: ArbNet provides the calibrated, certified surface; deep hedging provides the policy under the calibrated dynamics.

2.5. *Universal approximation in option pricing*

The lineage of universal-approximation theorems for neural networks runs from Cybenko (1989) and Hornik (1989; 1991) through Leshno et al. (1993), Pinkus (1999), and Yarotsky (2017). The classical results establish L^∞ density of multi-layer feed-forward networks in continuous functions on compacta. Less well-known is the shape-constrained version: ICNNs are dense in continuous convex functions (Amos et al., 2017; Chen et al., 2019; Bunne et al., 2022), and monotone networks are dense in continuous non-decreasing functions (Daniels and Velikova, 2010; Wehenkel and Louppe, 2019; Runje and Shankaranarayana, 2023; Guo and Lan, 2020). Sirignano and Spiliopoulos (2019) prove a deep Galerkin universal-approximation theorem in the context of partial differential equation solvers. Our Theorem 5.10 extends this lineage to the cone $\mathcal{C}_{\mathrm{cvx},\nearrow}$ of strike-convex, time-monotone, non-negative surfaces.

3. Preliminaries and notation

Throughout, $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t\geq 0}, \mathbb{Q})$ is a filtered probability space carrying a càdlàg semimartingale S that models the underlying asset under a risk-neutral measure $\mathbb{Q}$, with $S_0 = S > 0$. The risk-free rate r and dividend yield q are deterministic constants satisfying $r \geq q \geq 0$. We take the European call price $C(K,T) = e^{-rT}\mathbb{E}^{\mathbb{Q}}[(S_T - K)^+]$ as the primary modelling object on the strike-maturity domain $\mathbb{R}_{\geq 0} \times [0, T_{\max}]$ at a fixed valuation date $t = 0$, and recover implied volatility where needed by inversion of the Black–Scholes formula (Black and Scholes, 1973). Table 1 fixes symbols. The conventions for indices (i for data, j for mixture experts, ℓ for network layers) and the dual reading of K, T, S as scalars or batched 1-tensors are clear from context.

3.1. *Static no-arbitrage as a shape constraint on $C(\cdot,\cdot)$*

Static no-arbitrage — the impossibility of constructing, from the call cross-section at $t = 0$, a portfolio with non-negative future payoff and strictly negative cost today — imposes four pointwise shape constraints on $C(\cdot,\cdot)$. We list them in the form in which the architecture of Section 4 will enforce them.

Table 1. Notation used throughout.

Symbol	Meaning
S, K, T	Spot, strike, time-to-maturity (in years)
r, q	Risk-free rate, dividend yield ($r \geq q \geq 0$)
$F_T = S\, e^{(r-q)T}$	Forward at time T
$C(K, T)$	Call price at strike K, maturity T
$(F_T - K)^+$	Intrinsic value of forward call
$\partial_{KK} C$	Second strike derivative (proportional to risk-neutral density)
$\mathbb{Q}$	Risk-neutral measure
$\sigma(K, T)$	Implied volatility (Black–Scholes-inverted)
$w(k, T) = \sigma^2 T$	Total implied variance, $k = \log(K/F_T)$
θ	Parameter vector (network weights and gates)
Θ	Parameter space
$\mathrm{softplus}(u) = \log(1 + e^u)$	Smooth ReLU
$\omega_f(\delta)$	Modulus of continuity of f
J	Number of mixture experts in ArbNet
ϕ_j^{cvx}	jth ICNN (convex in strike input)
ϕ_j^{mon}	jth Mono (non-decreasing in time input)
$s_{j,\theta}, \alpha_\theta$	Per-expert scale gate; boundary-rate gate
$\mathcal{B}$	Compact strike-maturity box $[K_0, K_1] \times [0, T_{\max}]$
$\mathcal{C}_{\mathrm{cvx},\nearrow}$	Cone of continuous, strike-convex, time-monotone, non-negative surfaces
$\rho_T(K)$	Risk-neutral density at maturity T

Definition 3.1 (Static no-arbitrage)**.** A continuous map $C : \mathbb{R}_{\geq 0} \times [0, T_{\max}] \to \mathbb{R}_{\geq 0}$ is *static-arbitrage-free* if

(A1) $K \mapsto C(K, T)$ is convex on $\mathbb{R}_{\geq 0}$, for every $T \in [0, T_{\max}]$;
(A2) $T \mapsto e^{rT}\, C(K, T)$ is non-decreasing on $[0, T_{\max}]$, for every K;
(A3) $C(K, 0) = (S - K)^+$ for every K;
(A4) $C(K, T) \geq e^{-rT}(F_T - K)^+$ for every (K, T).

Each condition is the analytic shadow of a textbook portfolio argument, and reading the four together is the easiest way to see why they exhaust the structural part of static arbitrage. Strike convexity (A1) rules out the butterfly portfolio: at three strikes $K_1 < K_2 < K_3$ with $K_2 = (K_1 + K_3)/2$, the position long $C(K_1) + C(K_3)$ against short $2C(K_2)$ has non-negative terminal payoff, so it must have non-negative cost today, which is precisely convexity in K. The Breeden–Litzenberger identity $\partial_{KK}^2 C(K, T) = e^{-rT} \rho_T(K)$ gives the equivalent probabilistic reading (Breeden and Litzenberger, 1978): a concavity at any K is a strictly negative implied density there, and therefore not a probability law. Calendar monotonicity (A2) is the dual of the calendar spread: at fixed K, holding the longer-dated call against the shorter one with cash carried at r yields a non-negative terminal payoff between $T_1 < T_2$, forcing $e^{rT_2} C(K, T_2) \geq e^{rT_1} C(K, T_1)$ at $t = 0$. The expiry condition (A3) is the definitional identification of $C(\cdot, 0)$ with the call payoff, and the intrinsic bound (A4) follows from put-call parity, $C(K, T) - P(K, T) = e^{-rT}(F_T - K)$, combined with $P(K, T) \geq 0$.

The list is not exhaustive. The upper envelope $C(K, T) \leq S e^{-qT}$ and the strike-monotonicity $K \mapsto C(K, T)$ non-increasing, together with several less familiar conditions catalogued in (Cousot, 2007; Davis and Hobson, 2007; Roper, 2010), are derivable from (A1)–(A4) under mild regularity, and Section 5.2 verifies that they hold automatically for the surfaces our architecture produces. We confine the structural enforcement to (A1)–(A4) for two reasons. First, this is the canonical

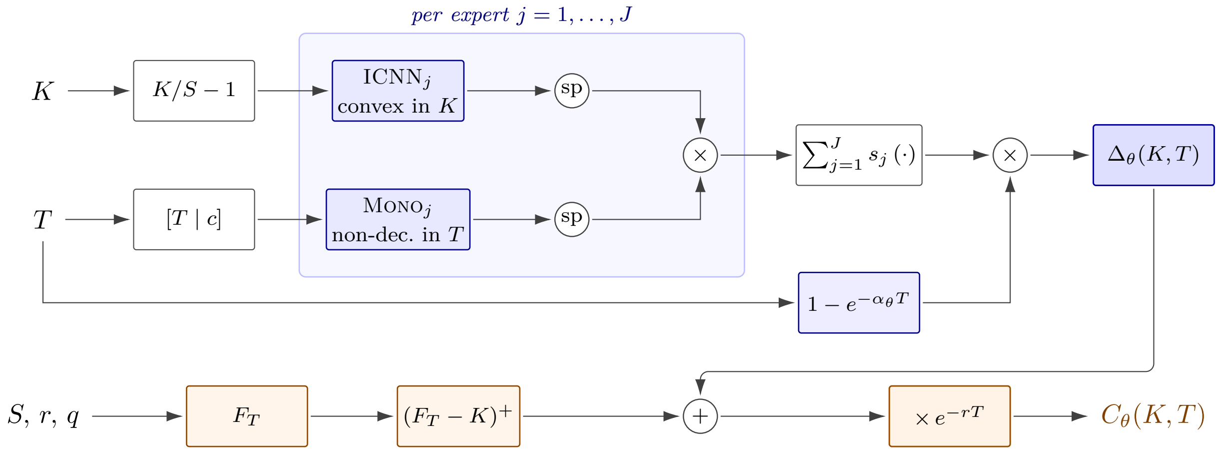

Figure 1. Computational graph of ArbNet. The K and T inputs feed two parallel trainable pathways — an input-convex network in K (top) and a monotone network in T (middle), shaded blue — whose softplus-rectified outputs combine into a per-expert product. The mixture sum $\sum_j s_j(\cdot)$ over J experts is multiplied by the smooth boundary factor $1 - e^{-\alpha_\theta T}$ to produce the time-value correction $\Delta_\theta(K,T)$. This is added to the forward-intrinsic baseline $(F_T - K)^+$ (bottom, shaded amber) and discounted by e^{-rT} to yield the call price. By Theorem 5.4, conditions (A1)–(A4) hold for every parameter setting.

list against which neural pricers have been scrutinised (Ackerer et al., 2020; Chataigner et al., 2020; Horvath et al., 2021; Ruf and Wang, 2020b), so reporting against it makes the comparison with prior work direct. Second, the four conditions admit an elementary architectural realisation — convexity, monotonicity, an exponential boundary factor, an additive intrinsic baseline — that the longer list does not. The price of this parsimony is a sharply delimited hypothesis class, which we make explicit and quantify in Theorem 5.11.

3.2. *The Lee moment formula: a structural condition of a different character*

A fifth condition governs the asymptotic shape of the smile rather than pointwise positivity. Lee's moment formula (Lee, 2004, 2005) couples the integrability of S_T under $\mathbb{Q}$ to the wing slope of the total implied variance $w(k,T) = \sigma^2(K,T)\,T$ in log-moneyness $k = \log(K/F_T)$: the right-wing slope β_R in $w(k,T) \sim \beta_R\, k$ as $k \to +\infty$ satisfies $\beta_R \le 2$, and the left wing obeys the symmetric bound $\beta_L \le 2$, with equality attained only when the corresponding moment of S_T fails to exist on $\mathbb{R}_+$. Unlike (A1)–(A4), this is an asymptotic inequality, not a pointwise one, and it is not preserved by the elementary convex-monotone compositions on which our construction rests. We therefore treat it as a monitored diagnostic rather than a structural target. Section 5.5 discusses when this matters and what light-touch mitigations (weight clamping, Lipschitz-constrained variants) are available; Section 8.1 reports the empirical wing behaviour on real NSE quotes.

4. The ArbNet architecture

4.1. *Computational graph*

The architecture is shown in Figure 1. The graph has two parallel pathways from inputs (K,T), a per-expert combiner, a mixture sum, a smooth boundary factor, and a final composition with the intrinsic baseline.

The architecture has a small number of explicit components. We list them and motivate each.

4.2. *Formal definitions of the building blocks*

4.2.1. Input-convex network (ICNN).

Fix depth L and widths $(w_0, \ldots, w_L)$ with $w_0 = 1$, $w_L = 1$. The input-convex recurrence of Amos et al. (2017) reads

$$z_{\ell+1} = \sigma\big(W_\ell^{(z)} z_\ell + W_\ell^{(x)} x + W_\ell^{(c)} c + b_\ell\big), \quad z_0 = 0,\ x = K/S, \tag{4}$$

where σ is a convex non-decreasing activation (we use softplus), c is an optional auxiliary context (left at zero in the baseline configuration), and the weight constraint

$$W_\ell^{(z)} \in \mathbb{R}_{\geq 0}^{w_{\ell+1} \times w_\ell} \tag{5}$$

is enforced by reparameterisation $W_\ell^{(z)} = \mathrm{softplus}(\widetilde{W}_\ell^{(z)})$ for unconstrained $\widetilde{W}_\ell^{(z)}$. The output is $\phi^{\mathrm{cvx}}(x; c) := z_L$. We use $L = 2$ and width 64 throughout. The recurrence has the key structural property:

Proposition 4.1 (ICNN convexity, (Amos et al., 2017))**.** *$x \mapsto \phi^{\mathrm{cvx}}(x; c)$ is convex on $\mathbb{R}$ for every c.*

The proof is by induction on ℓ: the affine layer $W_\ell^{(z)} z_\ell + W_\ell^{(x)} x + W_\ell^{(c)} c + b_\ell$ is convex in x if z_ℓ is convex (by the non-negativity constraint on $W_\ell^{(z)}$ and the convexity-preserving cone closure of Theorem 5.2); the activation σ is convex non-decreasing, so by Theorem 5.1 the composition is convex.

4.2.2. Monotone network (Mono).

Fix depth L' and widths $(w'_0, \ldots, w'_{L'})$ with $w'_0 = 1 + d_c$, $w'_{L'} = 1$. The monotone recurrence of Daniels and Velikova (2010); Sill (1998) reads

$$h_{\ell+1} = \sigma\big(V_\ell^{(m)} h_\ell^{(m)} + V_\ell^{(c)} h_\ell^{(c)} + d_\ell\big), \tag{6}$$

with $V_\ell^{(m)} \in \mathbb{R}_{\geq 0}^{w'_{\ell+1} \times w_\ell^{(m)}}$ (softplus-reparameterised) and $V_\ell^{(c)}$ unconstrained. The activation σ is non-decreasing (softplus). The output is $\phi^{\mathrm{mon}}(T; c) := h_{L'}$. We use $L' = 2$ and width 32 throughout. Recent extensions of monotone networks — including unconstrained monotonic neural networks (Wehenkel and Louppe, 2019), the Lipschitz monotonic networks of Runje and Shankaranarayana (2023), and the polynomial-spline-based monotone approximators of Guo and Lan (2020) — could be used as drop-in replacements; we use the classical Daniels–Velikova construction for simplicity.

Proposition 4.2 (Mono monotonicity, (Daniels and Velikova, 2010))**.** *$T \mapsto \phi^{\mathrm{mon}}(T; c)$ is non-decreasing on $\mathbb{R}_{\geq 0}$ for every c.*

4.2.3. Scalar gates.

The boundary growth rate $\alpha_\theta = \mathrm{softplus}(\widetilde{\alpha}) > 0$ is a learned positive scalar that controls the speed at which the time-value factor $1 - e^{-\alpha_\theta T}$ leaves zero. The per-expert positive scales $s_{j,\theta} = \mathrm{softplus}(\widetilde{s}_j) > 0$ control the relative weighting of each expert. Both are unconstrained-parameterised by softplus.

4.3. *Design rationale*

Several non-obvious choices in Equations (1) and (2) reward a brief look.

Algorithm 1 ArbNet forward pass

Require: strike vector K, maturity vector T, spot S, rate r, dividend yield q, parameters θ
Ensure: call prices C

$F \leftarrow S \cdot \exp((r-q)T); \quad I \leftarrow \max(F-K, 0)$
$x \leftarrow ((K/S)-1) \cdot \kappa; \quad u \leftarrow [T \,\|\, c]; \quad \Delta \leftarrow 0$
for $j = 1, \ldots, J$ **do**
  $f_j \leftarrow \mathrm{softplus}(\phi_j^{\mathrm{cvx}}(x))$ ▷ convex in K, ≥ 0
  $g_j \leftarrow \mathrm{softplus}(\phi_j^{\mathrm{mon}}(u))$ ▷ non-decreasing in T, ≥ 0
  $s_j \leftarrow \mathrm{softplus}(\widetilde{s}_j); \quad \Delta \leftarrow \Delta + s_j f_j g_j$
$\alpha \leftarrow \mathrm{softplus}(\widetilde{\alpha}); \quad \Delta \leftarrow S\,(1-\exp(-\alpha T))\,\Delta$
$C \leftarrow \exp(-rT) \cdot (I + \Delta)$ **return** C

The first is the additive decomposition $C_\theta = e^{-rT}[(F_T - K)^+ + \Delta_\theta]$ into a forward intrinsic and a non-negative correction. This is more than cosmetic. The intrinsic is the model-free no-arbitrage lower bound (A4), common to every static-arbitrage-free pricer; absorbing it into a fixed baseline leaves the network with the much smaller task of learning the *time value*, which is by construction non-negative on any compliant surface. Two practical benefits follow. The learning problem is more constrained: $\Delta_\theta \geq 0$ is a tighter target than the call price, which spans zero to deep-ITM intrinsic. And it is more numerically stable, because the dynamic range of the time value is far narrower than the dynamic range of C itself.

The second is the mixture-of-products structure $\Delta_\theta = \sum_{j=1}^{J} s_{j,\theta}\, f_j(K)\, g_j(T)$. A single product $f(K)g(T)$ would force separability, and empirical surfaces are emphatically not separable: the smile flattens with maturity, in line with the Lee asymptotic, and the at-the-money term structure has shape. A finite mixture is the minimal extension that breaks separability while preserving the structural compliance, since the cone of strike-convex, time-monotone, non-negative functions is closed under non-negative linear combinations. The choice of J is then a capacity dial, with $J \to \infty$ delivering density by Theorem 5.10.

The third is the boundary factor $1 - e^{-\alpha_\theta T}$. The expiry condition (A3), in the decomposition above, reduces to the requirement $\Delta_\theta(K, 0) = 0$ for every K. Multiplying the entire correction by a strictly non-decreasing function of T that vanishes at the origin enforces this without further restriction; the exponential form is the natural choice because it depends on a single learnable rate and has strictly positive derivative $\alpha_\theta e^{-\alpha_\theta T} > 0$ on the entire interval, which is what Theorem 5.3 requires to preserve monotonicity downstream.

The fourth is the use of softplus, $\mathrm{softplus}(u) = \log(1+e^u)$, throughout. Softplus is convex, non-decreasing, and strictly positive — the three properties the compositions of Theorems 5.1 and 5.3 need — and its strict positivity matters in practice as well as in theory: gradients propagate through the per-expert product even when the inner ICNN output is small, which stabilises early training. ReLU would zero out gradients on a measure-zero set in principle but on a non-trivial fraction in finite-precision arithmetic, and gives worse initialisation behaviour for the inner ICNNs.

The fifth, and most consequential, is the decision to work in the call-price domain rather than the implied-volatility domain. This deserves a separate discussion and we give it in Section 4.5.

4.4. *Forward pass and computational complexity*

Algorithm 1 gives the forward pass in pseudocode.

The total parameter count is $J \cdot (P_{\mathrm{ICNN}} + P_{\mathrm{MONO}}) + J + 1$, where P_{ICNN} and P_{MONO} are the per-network parameter counts. For our default configuration ($J = 6$, ICNN $[64, 64]$, MonoNet $[32, 32]$), this evaluates to approximately 33,000 parameters in the context-free setting, rising to approximately 43,000 when a per-day context vector feeds the ICNN and monotone modules, as in the real-data study of Section 7.2. The forward-pass cost for a batch of size B is $\mathcal{O}(B \cdot J \cdot (\mathrm{width}_{\mathrm{ICNN}}^2 L + \mathrm{width}_{\mathrm{MONO}}^2 L'))$, which is asymptotically identical to a soft-penalty MLP; the structural constraints add no computational overhead, because the softplus reparameterisation of the ICNN's $W_\ell^{(z)}$ matrices is a fixed per-layer cost that does not scale with the batch size.

4.5. *Why the call-price domain rather than the implied-vol domain*

A natural alternative to our construction would parameterise the implied total variance $w_\theta(k,T)$ directly using ICNN/MonoNet building blocks, with $k = \log(K/F_T)$. The conversion from implied variance to call price is via the Black–Scholes formula and is therefore one-to-one. However, the static-arbitrage conditions in the implied-vol domain are not preserved by simple compositions. The Durrleman positivity (Gatheral and Jacquier, 2014a,b) reads

$$g(k,T) := \left(1 - \frac{k\,\partial_k w}{2w}\right)^2 - \frac{(\partial_k w)^2}{4}\left(\frac{1}{w} + \frac{1}{4}\right) + \frac{\partial_k^2 w}{2} \geq 0, \tag{7}$$

which is a non-linear functional of w and its first two derivatives. Convexity of w in k implies the third term is non-negative, but does not imply the whole expression is non-negative; one can produce convex w for which $g(k,T) < 0$ at some k. The implied-vol-domain construction therefore reduces to either (a) imposing the Durrleman condition as a soft penalty (recovering the soft-penalty paradigm) or (b) restricting to parameterisations like SVI where the Durrleman condition is implied by simpler parameter constraints, which limits expressivity.

In the call-price domain, by contrast, the no-arbitrage conditions (A1)–(A4) are first-order linear (convexity in K, monotonicity in T of $e^{rT}C$). These properties are preserved by the elementary compositions used in Equation (2) (non-negative weighted sums, products of non-negative monotone functions, composition with convex non-decreasing scalar functions). The call-price domain therefore admits a structural neural pricer with simple primitives; the implied-vol domain does not.

4.6. *Greeks via automatic differentiation*

Once the architecture is trained, all of the standard Greeks are computed by automatic differentiation (Baydin et al., 2018) of C_θ with respect to its inputs, using PyTorch's autograd (Paszke et al., 2019). Specifically:

- Delta $\delta(K,T) = \partial_S C_\theta$ (single backward pass).
- Gamma $\gamma(K,T) = \partial_K^2 C_\theta = e^{-rT}\rho_T(K)$ (double backward pass). Structurally non-negative by Theorem 5.4 (A1).
- Theta $\Theta(K,T) = -\partial_T C_\theta$ (single backward pass).
- Vega $\nu(K,T) = \partial_\sigma C_\theta\big|_{\sigma=\widehat{\sigma}(K,T)}$ (requires Newton inversion to obtain $\widehat{\sigma}$; see Section C).

The non-negativity of γ is a guarantee *absent* from soft-penalty pricers; it means that the delta is non-decreasing in strike (deeper out-of-the-money calls have smaller delta than at-the-money ones), which is a desirable monotonicity property for hedging stability. We discuss this further in Section 8.5.

5. Theoretical guarantees

5.1. *Auxiliary lemmas*

We collect three elementary lemmas of convex analysis (Boyd and Vandenberghe, 2004); their roles in the main theorem are flagged.

Lemma 5.1 (Convexity composition). *If $f : \mathbb{R} \to \mathbb{R}$ is convex and $h : \mathbb{R} \to \mathbb{R}$ is convex non-decreasing, then $h \circ f$ is convex.*

Proof. For $\lambda \in [0,1]$ and $x, y \in \mathbb{R}$,

$$\begin{aligned} h(f(\lambda x + (1-\lambda)y)) &\le h(\lambda f(x) + (1-\lambda)f(y)) && \text{(convexity of } f\text{, monotonicity of } h\text{)} \\ &\le \lambda h(f(x)) + (1-\lambda)h(f(y)) && \text{(convexity of } h\text{).} \end{aligned}$$

□ □

Lemma 5.2 (Cone closure of convex functions)**.** *Convex functions on a convex domain are closed under non-negative linear combinations and scalar multiplication by non-negative scalars. Formally, if $f_1, \ldots, f_N$ are convex and $\lambda_1, \ldots, \lambda_N \ge 0$, then $\sum_i \lambda_i f_i$ is convex.*

Lemma 5.3 (Monotone product)**.** *If u, v are non-negative non-decreasing functions on $D \subseteq \mathbb{R}$, then uv is non-decreasing on D. In particular, for any non-negative non-decreasing $u_1, \ldots, u_N$, $\prod_i u_i$ is non-decreasing.*

Proof. For $x \le y$,

$$u(y)v(y) - u(x)v(x) = u(y)\big(v(y) - v(x)\big) + v(x)\big(u(y) - u(x)\big) \ge 0,$$

by non-negativity and monotonicity of u and v. The general statement follows by induction. □
□

5.2. *Structural arbitrage-freeness*

Theorem 5.4 (Arbitrage-freeness of ArbNet)**.** *Let C_θ be defined by Equations* (1) *and* (2) *with ϕ_j^{cvx} satisfying Equations* (4) *and* (5)*, ϕ_j^{mon} satisfying Equation* (6)*, and $\alpha_\theta, s_{j,\theta} > 0$. Assume $r \ge q \ge 0$. Then conditions* (A1)–(A4) *of Theorem 3.1 hold for every $\theta \in \Theta$.*

Proof. We prove each condition in turn.

5.2.1. Condition (A1)*: convexity in strike.*

We show that $K \mapsto C_\theta(K,T)$ is convex on $\mathbb{R}$ for every fixed $T \ge 0$.

The forward intrinsic $(F_T - K)^+ = \max(F_T - K, 0)$ is convex in K as the pointwise maximum of two affine functions; multiplication by the positive constant e^{-rT} preserves convexity (Theorem 5.2).

For the correction $\Delta_\theta(K,T)$, we need to show that $K \mapsto \Delta_\theta(K,T)$ is convex for every fixed T. The strike enters only through the inner ICNN compositions $\phi_j^{\mathrm{cvx}}(K/S)$. By Theorem 4.1, $u \mapsto \phi_j^{\mathrm{cvx}}(u)$ is convex on $\mathbb{R}$. The affine substitution $u = K/S$ preserves convexity, so $K \mapsto \phi_j^{\mathrm{cvx}}(K/S)$ is convex. Softplus is convex non-decreasing, so by Theorem 5.1, $K \mapsto \mathrm{softplus}(\phi_j^{\mathrm{cvx}}(K/S))$ is convex.

For each j, the time-dependent factor $\mathrm{softplus}(\phi_j^{\mathrm{mon}}(T))$ is a constant in K that is non-negative. The per-expert scale $s_{j,\theta}$ is non-negative. By Theorem 5.2, the per-expert contribution

$$K \mapsto s_{j,\theta}\ \mathrm{softplus}(\phi_j^{\mathrm{cvx}}(K/S))\ \mathrm{softplus}(\phi_j^{\mathrm{mon}}(T))$$

is a non-negative scalar multiple of a convex function in K, hence convex in K. The mixture sum over j is a non-negative linear combination of convex functions, hence convex (Theorem 5.2). Multiplying by the constants S and $1 - e^{-\alpha_\theta T}$ (both non-negative) preserves convexity. Therefore $K \mapsto \Delta_\theta(K,T)$ is convex.

Finally, $K \mapsto C_\theta(K,T) = e^{-rT}[(F_T - K)^+ + \Delta_\theta(K,T)]$ is a non-negative linear combination of two convex functions in K, hence convex.

5.2.2. Condition (A2)*: carry-adjusted calendar monotonicity.*

We show that $T \mapsto e^{rT} C_\theta(K,T)$ is non-decreasing on $[0, T_{\max}]$ for every fixed $K \geq 0$. From Equation (1),

$$e^{rT} C_\theta(K,T) = (F_T - K)^+ + \Delta_\theta(K,T).$$

For the intrinsic, $T \mapsto F_T = Se^{(r-q)T}$ is non-decreasing under the assumption $r \geq q$; consequently $T \mapsto F_T - K$ is non-decreasing, and the positive-part operator $z \mapsto z^+$ is non-decreasing, so $T \mapsto (F_T - K)^+$ is non-decreasing.

For the correction, the strike-dependent factor $\mathrm{softplus}(\phi_j^{\mathrm{cvx}}(K/S))$ is a non-negative constant in T. By Theorem 4.2, $T \mapsto \phi_j^{\mathrm{mon}}(T)$ is non-decreasing, and softplus is non-decreasing, so $T \mapsto \mathrm{softplus}(\phi_j^{\mathrm{mon}}(T))$ is non-decreasing. The product

$$T \mapsto \mathrm{softplus}(\phi_j^{\mathrm{cvx}}(K/S))\ \mathrm{softplus}(\phi_j^{\mathrm{mon}}(T))$$

is the product of a non-negative constant in T and a non-negative non-decreasing function of T, hence non-negative non-decreasing in T (this is the degenerate case of Theorem 5.3). Multiplying by $s_{j,\theta} > 0$ and summing over j preserves non-negative non-decreasingness. Multiplying by the non-negative non-decreasing boundary factor $1 - e^{-\alpha_\theta T}$ (whose derivative is $\alpha_\theta e^{-\alpha_\theta T} > 0$) preserves non-decreasingness by Theorem 5.3. Multiplying by the positive constant S preserves it. Therefore $T \mapsto \Delta_\theta(K,T)$ is non-decreasing.

Summing the two non-decreasing functions yields a non-decreasing function.

5.2.3. Condition (A3)*: expiry boundary.*

At $T = 0$: $F_0 = Se^{(r-q)\cdot 0} = S$, so $(F_0 - K)^+ = (S-K)^+$. The boundary factor is $1 - e^{-\alpha_\theta \cdot 0} = 0$, so $\Delta_\theta(K,0) = 0$. Hence $C_\theta(K,0) = e^0[(S-K)^+ + 0] = (S-K)^+$.

5.2.4. Condition (A4)*: forward intrinsic lower bound.*

Every factor in the expression for Δ_θ is non-negative: $S > 0$; $1 - e^{-\alpha_\theta T} \geq 0$ for $T \geq 0$; the $s_{j,\theta}$ are positive; both softplus terms are positive. Therefore $\Delta_\theta(K,T) \geq 0$, and

$$C_\theta(K,T) = e^{-rT}[(F_T - K)^+ + \Delta_\theta(K,T)] \geq e^{-rT}(F_T - K)^+,$$

which is the desired bound. □ □

Corollary 5.5 (Implied risk-neutral density)**.** *For every $T > 0$ and $\theta \in \Theta$, the implied risk-neutral density of S_T under $\mathbb{Q}$, defined by Breeden–Litzenberger as*

$$\rho_T^\theta(K) := e^{rT} \partial_K^2 C_\theta(K,T),$$

is a non-negative Borel measure on $\mathbb{R}_{\geq 0}$. It decomposes as

$$\rho_T^\theta \ = \ \delta_{F_T} \ + \ e^{rT}\, \partial_K^2 \Delta_\theta(\cdot, T),$$

a Dirac atom at the forward $K = F_T$, inherited from the kink of the intrinsic $(F_T - K)^+$, plus an absolutely continuous part with non-negative Lebesgue density $e^{rT}\partial_K^2 \Delta_\theta \geq 0$. Because the correction Δ_θ is everywhere twice differentiable, it cannot cancel the intrinsic kink: the atom persists for every *$T > 0$.* ArbNet *therefore assigns strictly positive risk-neutral probability to the event $\{S_T = F_T\}$. This is a genuine and inspectable modelling artifact of the call-domain construction — it appears as the density spike in Figure 5(d) — and we revisit it among the limitations of Section 9.3.*

5.3. *Universal approximation*

Definition 5.6 (Convex-monotone class)**.** Let $\mathcal{C}_{\mathrm{cvx},\nearrow}(\mathcal{B})$ be the set of continuous functions $g : \mathcal{B} = [K_0, K_1] \times [0, T_{\max}] \to \mathbb{R}_{\geq 0}$ that are strike-convex (*i.e.*, $K \mapsto g(K,T)$ is convex for every T), time-non-decreasing (*i.e.*, $T \mapsto g(K,T)$ is non-decreasing for every K), and satisfy $g(K,0) = 0$ for every K.

Lemma 5.7 (Tensor-product density with explicit rate)**.** *Let $g \in \mathcal{C}_{\mathrm{cvx},\nearrow}(\mathcal{B})$ and $\varepsilon_1 > 0$. Set $\delta = \omega_g^{-1}(\varepsilon_1)$ (the largest δ such that $\omega_g(\delta) \leq \varepsilon_1$, where ω_g is the bivariate modulus of continuity of g). Let $\{\kappa_\ell\}_{\ell=0}^{M_K}$ and $\{\tau_n\}_{n=0}^{M_T}$ be uniform partitions of $[K_0, K_1]$ and $[0, T_{\max}]$ with $M_K = \lceil (K_1 - K_0)/\delta \rceil$ and $M_T = \lceil T_{\max}/\delta \rceil$. Then there exist non-negative coefficients $\lambda_{\ell,n} \geq 0$ such that*

$$g_M(K,T) := \sum_{\ell=0}^{M_K} \sum_{n=0}^{M_T} \lambda_{\ell,n} \, (K - \kappa_\ell)^+ \, (T - \tau_n)^+ \tag{8}$$

satisfies $\|g - g_M\|_\infty \leq 3\varepsilon_1$ on $\mathcal{B}$.

The proof, deferred to Section A.1, has three steps: piecewise-linear interpolation of $\tau \mapsto g(K,\tau)$ at the breakpoints $\{\tau_n\}$ produces non-negative slopes $a_n(K) \geq 0$; piecewise-linear convex interpolation of $K \mapsto a_n(K)$ produces non-negative slopes $\beta_{n,\ell} \geq 0$; combining gives the rank-one decomposition.

Lemma 5.8 (ICNN density on convex non-negative functions)**.** *For any continuous convex function $u : [a,b] \to \mathbb{R}_{\geq 0}$ and any $\eta > 0$, there exists an ICNN ϕ as in Equations* (4) *and* (5) *with softplus activations such that $\|u - \mathrm{softplus}(\phi)\|_\infty < \eta$.*

Sketch: this is a corollary of the input-convex universal approximation theorem of Amos–Xu (2017), with subsequent refinements in (Chen et al., 2019; Bunne et al., 2022). ICNNs with softplus activations are L^∞-dense in continuous convex functions on a compact interval; composing with $\mathrm{softplus}(\widehat{\phi} - \log 2)$ produces the non-negative output.

Lemma 5.9 (Mono density with boundary)**.** *For any continuous non-decreasing function $v : [0, T'] \to \mathbb{R}_{\geq 0}$ with $v(0) = 0$ and any $\eta > 0$, there exist $\alpha > 0$ and a monotone network ψ with $\left\| v(T) - (1 - e^{-\alpha T}) \, \mathrm{softplus}(\psi(T)) \right\|_\infty < \eta$ on $[0, T']$.*

Sketch: Daniels and Velikova (2010) establish L^∞-density of monotone networks on continuous non-decreasing functions on a compact interval. The boundary factor $1 - e^{-\alpha T}$ with α large enough approximates the unit step at $T = 0$, removing the boundary singularity in v. Combining gives the result; full details in Section A.2.

Theorem 5.10 (Universal approximation with explicit rate)**.** *Let $g \in \mathcal{C}_{\mathrm{cvx},\nearrow}(\mathcal{B})$ and $\varepsilon > 0$. Then there exists an ArbNet correction $\widehat{g}$ of the form Equation* (2) *with $\|g - \widehat{g}\|_\infty < \varepsilon$. Moreover, the number of mixture components required is $J = \mathcal{O}(\omega_g^{-1}(\varepsilon/3)^{-2})$.*

Proof. Apply Theorem 5.7 with $\varepsilon_1 = \varepsilon/9$ to obtain g_M with $\|g - g_M\|_\infty \leq \varepsilon/3$ and $J = (M_K + 1)(M_T + 1) \leq C\omega_g^{-1}(\varepsilon/9)^{-2}$. Each rank-one term $\lambda_{\ell,n}(K - \kappa_\ell)^+(T - \tau_n)^+$ has a strike factor $(K - \kappa_\ell)^+$ that is continuous convex non-negative on $[K_0, K_1]$, and a time factor $(T - \tau_n)^+$ that is continuous non-decreasing non-negative on $[0, T_{\max}]$ with value zero at $T = 0$.

Apply Theorem 5.8 to approximate each $(K - \kappa_\ell)^+$ to within $\eta = \varepsilon/(3J\overline{uv})$ in L^∞, where $\overline{uv}$ is the maximum of $u_\ell v_n$ over the partition (a finite constant depending on g and $\mathcal{B}$). Apply Theorem 5.9 with a shared α to approximate each $(T - \tau_n)^+$ to within the same η. The product of two approximations is within $\eta \cdot \max + \max \cdot \eta = 2\eta \cdot \max$ of the true product, and summing J such terms with non-negative coefficients $\lambda_{\ell,n}$ keeps the bound at $\|g_M - \widehat{g}\|_\infty \leq \varepsilon/3$ by triangle inequality.

Combining via triangle inequality, $\|g - \widehat{g}\|_\infty \leq \|g - g_M\|_\infty + \|g_M - \widehat{g}\|_\infty \leq 2\varepsilon/3 < \varepsilon$. $\square$ $\square$

5.4. *Expressivity limit and the BS exclusion*

Theorem 5.10 states that the architecture is universal on the class $\mathcal{C}_{\mathrm{cvx},\nearrow}$. It is, however, *not universal* on the strictly larger class of all arbitrage-free call surfaces, because not every arbitrage-free call surface has a convex time-value in K. The most striking instance is the Black–Scholes family.

Proposition 5.11 (Black–Scholes surfaces are not representable)**.** *Let $C^{\mathrm{BS}}(K, T; \sigma)$ be the Black–Scholes call price with constant volatility $\sigma > 0$ and rates $r \geq q \geq 0$. Then $C^{\mathrm{BS}}(\cdot, \cdot; \sigma)$ does not lie in the image of the parameterisation Equations* (1) *and* (2) *for any $\theta \in \Theta$.*

Proof. The forward time value of Black–Scholes is

$$\widetilde{\Delta}^{\mathrm{BS}}(K, T; \sigma) := e^{rT} C^{\mathrm{BS}}(K, T; \sigma) - (F_T - K)^+.$$

This quantity is the difference between two quantities: (i) the carry-adjusted call price, which is smooth and strictly increasing in T, and (ii) the intrinsic, which has a non-differentiable kink at $K = F_T$. At $K = F_T$, the kink contribution makes $\widetilde{\Delta}^{\mathrm{BS}}$ attain its global maximum in K (because the time value is maximised at-the-money under Black–Scholes for any $T > 0$ and $\sigma > 0$, a standard textbook result (Hull, 2018)).

A function that attains a strict local maximum on an open set cannot be convex on that set. Therefore $K \mapsto \widetilde{\Delta}^{\mathrm{BS}}(K, T; \sigma)$ is non-convex on any open neighbourhood of $K = F_T$.

By Theorem 5.4 (A1), an ArbNet-realised correction $\Delta_\theta(K, T)$ is convex in K for every θ. Two functions that differ in their convexity class cannot coincide on an open set. Therefore the parameterisation cannot represent Black–Scholes. □ □

Proposition 5.11 is the structural cost of the architectural compliance: by hard-coding convexity in K into the architecture, we exclude any surface whose forward time-value is non-convex in K. This includes the entire Black–Scholes family, and more generally any surface where the time-value is bell-shaped (peaking at-the-money and decaying in both directions). Rough-Bergomi surfaces have less pronounced bell shapes than Black–Scholes at short maturities but still some; the empirical signature of this restriction appears in Figure 5, where ArbNet's risk-neutral density is over-concentrated at the money.

It is worth noting that no *soft-penalty* pricer has this exclusion: the soft-penalty MLP can represent any surface, just not without residual violations. The exclusion is therefore the explicit cost of the architectural-vs-penalty trade.

5.5. *The Lee wing slope*

A separate observation concerns the asymptotic shape of the smile. The Lee moment formula (Lee, 2004, 2005), recalled in Section 3.2, requires the right-wing total variance to grow at most linearly in log-moneyness, with slope at most 2. Our parameterisation does not enforce this bound structurally: in principle, the ICNN ϕ_j^{cvx} can produce a quadratic or super-linear growth in K/S, which translates to a super-linear growth in $w(k, T)$ at large $|k|$. In practice, we observe that even untrained ICNNs (initialised with small weights) produce slopes well below 2, and trained ICNNs on data covering $\pm 30\%$ moneyness remain bounded. For training on data covering deeper wings ($\pm 50\%$ or more), one can mitigate by soft clamping of the last-layer ICNN weights, or by penalising the empirical wing slope with a weak regulariser. We do not pursue this in the present study because our training data is confined to the $\pm 30\%$ moneyness band where the issue does not arise.

6. Training objective and algorithms

6.1. *Loss function*

Given a batch $\mathcal{D} = \{(K_i, T_i, C_i^{\text{obs}}, c_i)\}_{i=1}^N$ of strike-maturity quotes (with optional context c_i), we define the total training loss as

$$\mathcal{L}_{\text{total}}(\theta) = \mathcal{L}_{\text{price}}(\theta) + \lambda_{\text{IV}}\, \mathcal{L}_{\text{IV}}(\theta) + \lambda_{\text{hedge}}\, \mathcal{L}_{\text{hedge}}(\theta), \tag{9}$$

where the three terms are the vega-weighted price RMSE, the implied-volatility RMSE (optional), and the deep-hedging $\text{CVaR}_{0.95}$ (optional). Crucially, *no soft no-arbitrage penalty is required*: the parameterisation is structurally compliant by Theorem 5.4. The Ackerer-style baseline used for comparison adds a penalty term $\lambda_{\text{arb}}\mathcal{L}_{\text{arb}}$, while ArbNet sets $\lambda_{\text{arb}} = 0$.

6.2. *Vega-weighted price loss*

The price loss is

$$\mathcal{L}_{\text{price}}(\theta) = \frac{1}{N} \sum_{i=1}^{N} \frac{(C_\theta(K_i, T_i) - C_i^{\text{obs}})^2}{\max(\nu(K_i, T_i)^2, w_{\min}^2)}, \tag{10}$$

where $\nu(K_i, T_i)$ is the Black–Scholes vega evaluated at the observed implied volatility σ_i^{obs} and $w_{\min} > 0$ is a floor. The vega weighting reflects the standard market-practitioner observation that pricing errors in the wings (where ν is small) are much smaller than pricing errors near the money in absolute terms but comparable in volatility space; weighting by vega-squared converts the price RMSE into a quantity that is approximately proportional to the implied-volatility RMSE. The floor $w_{\min}$ prevents the loss from blowing up at deep-OTM quotes where vega is essentially zero; we use $w_{\min} = 10^{-3}$ throughout.

6.3. *Implied volatility loss (optional)*

The implied-volatility loss is

$$\mathcal{L}_{\text{IV}}(\theta) = \frac{1}{N} \sum_{i=1}^{N} \left(\widehat{\sigma}_\theta(K_i, T_i) - \sigma_i^{\text{obs}}\right)^2, \tag{11}$$

where $\widehat{\sigma}_\theta$ is the Newton-inverted Black–Scholes implied volatility of the model price $C_\theta(K_i, T_i)$. The inversion is detailed in Section C. We disable the IV loss ($\lambda_{\text{IV}} = 0$) in our experiments for two reasons: (i) the Newton inversion is unstable at deep wings, producing NaN gradients that propagate; and (ii) the vega-weighted price loss already approximates the IV-domain loss to first order. Empirically, enabling $\lambda_{\text{IV}} > 0$ does not improve out-of-sample IV RMSE materially on rough-Bergomi surfaces and substantially increases training instability.

6.4. *Deep-hedging loss (optional)*

The deep-hedging loss (Buehler et al., 2019; Chakrabarti et al., 2019; Imaki et al., 2021) simulates discrete delta hedging on Monte-Carlo paths under a market simulator (e.g., a calibrated rough-Bergomi process), and minimises a convex risk measure of the terminal hedged PnL:

$$\mathcal{L}_{\text{hedge}}(\theta) = \text{CVaR}_{0.95}\left[-\Pi_\theta^{(p)}\right], \tag{12}$$

Algorithm 2 One epoch of ArbNet training

Require: dataset $\mathcal{D}$, batch size B, learning rate η, gradient clip γ_{clip}, parameters θ

for each mini-batch $\mathcal{B} \subset \mathcal{D}$ **do**
 $C \leftarrow \text{ArbNet}_\theta(\cdot)$ via Algorithm 1
 $\mathcal{L} \leftarrow \mathcal{L}_{\text{price}} + \lambda_{\text{IV}}\mathcal{L}_{\text{IV}} + \lambda_{\text{hedge}}\mathcal{L}_{\text{hedge}}$
 $g \leftarrow \nabla_\theta \mathcal{L}$, $g \leftarrow g \cdot \min(1, \gamma_{\text{clip}}/\|g\|_2)$
 $\theta \leftarrow \text{AdamStep}(\theta, g, \eta)$
return θ

Algorithm 3 Scale-aware static-arbitrage diagnostic

Require: pricer $\mathcal{M}$, strike grid $K[1{:}n_K]$, maturity grid $T[1{:}n_T]$, (S, r, q), tolerance τ_{tol}
Ensure: $(n_{\text{bf}}, n_{\text{cal}})$, the violation counts

$C[i,j] \leftarrow \mathcal{M}(K_j, T_i, S, r, q)$ for all i, j
$n_{\text{bf}}, n_{\text{cal}} \leftarrow 0, 0$
for $i \in [1, n_T], j \in [2, n_K - 1]$ **do**
 $\Delta^2 \leftarrow (C_{i,j+1} - 2C_{i,j} + C_{i,j-1})/(\Delta K)^2$
 if $-\Delta^2 > \tau_{\text{tol}}$ **then** $n_{\text{bf}} \leftarrow n_{\text{bf}} + 1$
for $i \in [1, n_T - 1], j \in [1, n_K]$ **do**
 if $e^{rT_i}C_{i,j} - e^{rT_{i+1}}C_{i+1,j} > \tau_{\text{tol}}$ **then** $n_{\text{cal}} \leftarrow n_{\text{cal}} + 1$
return $(n_{\text{bf}}, n_{\text{cal}})$

where $\Pi_\theta^{(p)}$ is the PnL on path p when hedging using $\partial_S C_\theta$ as the delta. The deep-hedging loss is more expensive to evaluate than the price loss because each minibatch requires a forward simulation of paths and accumulation of incremental hedges. We use $\lambda_{\text{hedge}} = 0$ in our experiments and only compute the hedging diagnostic at evaluation time (Algorithm 4).

6.5. *Optimisation algorithm*

We use Adam (Kingma and Ba, 2015) with learning rate 1×10^{-2}, batch size 64, gradient clipping at ℓ_2 norm 5.0, and weight decay 10^{-6} for ArbNet; the same hyperparameters for the Ackerer baseline; and learning rate 2×10^{-2} for the Black–Scholes baseline (a single-parameter model that converges faster). All weights are initialised with the Glorot/He scheme (He et al., 2015) applied to the underlying unconstrained matrices $\widetilde{W}_\ell^{(z)}, V_\ell^{(m)}$, etc.; the softplus reparameterisation then projects into the constrained subspace.

6.6. *Static-arbitrage diagnostic*

For evaluation, we report butterfly and calendar violation rates with a scale-aware tolerance $\tau_{\text{tol}} = \max(10^{-7}, 10^{-6} \cdot \bar{C})$, where $\bar{C}$ is the mean call price across the diagnostic grid. Two precautions keep the diagnostic free of numerical artifacts. First, the diagnostic grid is evaluated in *double* (float64) precision: in single precision, second differences of 5,000-INR call prices can flag spurious sign changes of magnitude 10^{-5} INR on a structurally compliant surface, which would otherwise contaminate the violation counts. Second, the scale-aware tolerance discards any residual round-off below economic significance — a violation of magnitude 10^{-8} INR on a 5,000-INR call is an artifact of finite-precision arithmetic, not an arbitrage. The diagnostic is reported in Algorithm 3.

6.7. *Delta-hedging diagnostic*

The hedging diagnostic, Algorithm 4, simulates discrete delta-hedging on rough-Bergomi paths and reports the standard deviation and $\text{CVaR}_{0.95}$ of the terminal hedged PnL. The standard

Algorithm 4 Delta-hedge PnL simulation

Require: pricer $\mathcal{M}$, paths $\{S_t^{(p)}\}$, strike K, maturity T_{hedge}, implied vol $\widehat{\sigma}$, step dt, cost τ

for $p = 1, \dots, P$ **do**
  $V_0 \leftarrow \mathcal{M}(K, T_{\text{hedge}}, S_0^{(p)})$; $\delta_0 \leftarrow \partial_S \mathcal{M}(\cdot)$
  $\text{cash}_0 \leftarrow V_0 - \delta_0 S_0^{(p)}$
  for $t = 1, \dots, n_t$ **do**
    $\delta_t \leftarrow \partial_S \mathcal{M}(K, T_{\text{hedge}} - t\,dt, S_t^{(p)})$
    $\text{cash}_t \leftarrow e^{r\,dt}\text{cash}_{t-1} - (\delta_t - \delta_{t-1})S_t^{(p)} - \tau|\delta_t - \delta_{t-1}|S_t^{(p)}$
  $\Pi^{(p)} \leftarrow \text{cash}_{n_t} + \delta_{n_t} S_{n_t}^{(p)} - (S_{n_t}^{(p)} - K)^+$
return $\{\Pi^{(p)}\}$

deviation is a measure of hedging precision; the $\text{CVaR}_{0.95}$ is a tail measure capturing the worst-case PnL over the worst 5% of paths (Glasserman, 2003, 2012).

7. Experimental setup

7.1. *Synthetic surface generation*

We generate synthetic call surfaces from a rough-Bergomi process (Bayer et al., 2016, 2023). The rough-Bergomi model is a path-dependent stochastic volatility model in which the log-volatility is driven by a fractional Brownian motion of Hurst parameter $H < 1/2$; the corresponding instantaneous variance process is the exponential of a centred Gaussian process with covariance reflecting the roughness of the volatility paths. The model is calibrated to capture three stylised facts of equity-index volatility surfaces: (i) ATM volatility skew that decays with maturity at a rate $T^{H-1/2}$ for $H < 1/2$, which empirically matches the observed short-end behaviour; (ii) a non-zero wing slope at short maturities, captured by the ρ parameter; (iii) an ATM term-structure that decays towards a long-run level, captured by the initial variance curve $\xi_0(\cdot)$.

For our experiments we use a constant initial variance $\xi_0(t) = \xi_0$. The rough-Bergomi parameters are drawn from

$$\begin{aligned} H &\in [0.07, 0.20] && \text{(roughness)} \\ \eta &\in [1.0, 3.0] && \text{(vol-of-vol)} \\ \rho &\in [-0.9, -0.3] && \text{(spot-vol correlation)} \\ \xi_0 &\in [0.02, 0.10] && \text{(initial variance).} \end{aligned}$$

Each draw defines one surface. We sample 10 distinct surfaces from this grid. For each surface, we use the hybrid scheme of Bennedsen, Lunde, and Pakkanen (2017) to simulate 4,000 Monte-Carlo paths of the rough-Bergomi process. We then price European calls on a grid of 20 strikes spanning $\pm 30\%$ moneyness in 5 maturities $(7, 14, 30, 60, 90)$ days, for a total of 100 quotes per surface. The spot is $S = 20{,}000$ (matching Nifty-50 scale in INR), the risk-free rate is $r = 6.5\%$ (matching the Indian repo rate of mid-2024), and the dividend yield is $q = 1.2\%$ (matching the Nifty-50 trailing yield).

For each surface, we run three independent training seeds. The total number of (surface, seed) trials is therefore 30.

7.2. *Real NSE Nifty 50 option data*

The real-data study runs a day-by-day walk-forward over the National Stock Exchange of India (NSE) Nifty 50 index-option market.

7.2.1. *Primary data.*

The training data is the NSE Futures & Options daily *bhavcopy*: one file per trading day giving the settlement price, open interest, and volume of every listed contract. We retain the Nifty 50 index calls and take the official settlement price as the quote. The sample spans 1,359 trading days, 1 January 2019 to 5 July 2024, and therefore covers the COVID-19 crash of early 2020 and the post-2022 monetary-tightening cycle. The underlying spot S is the official Nifty 50 daily close; the risk-free rate r is the cut-off yield of the most recent 91-day Government of India Treasury-bill auction on or before each date; the dividend yield q is the NSE Nifty 50 factsheet yield. Every series is committed to the paper's repository, so the study runs with no network access.

7.2.2. *Quality filtering.*

Deep-out-of-the-money weekly options on NSE frequently carry stale settlement prices after volatility spikes. Before fitting, each daily cross-section is passed through a fixed quality filter that drops contracts with a settlement price below 0.05 INR, with $|\log(K/F)| > 0.5$, or that violate the elementary no-arbitrage price bounds. Per-contract implied volatilities are then obtained from the surviving quotes by the vectorised Newton inversion of Section C. No trading day in the sample is discarded by the pipeline: all 1,359 days are fitted and evaluated.

7.2.3. *Context features.*

Because ArbNet is arbitrage-free for *every* context vector (Theorem 5.4), it is safe to condition the ICNN and monotone modules on auxiliary market state. We assemble a ten-dimensional per-day context vector: India VIX, an at-the-money Bank Nifty implied volatility computed on the fly from the same bhavcopy, one-week realised volatility of the Nifty 50, the USD/INR rate, foreign- and domestic-institutional net cash-market flows, headline CPI and WPI inflation, IIP growth, and the number of calendar days to the next scheduled macroeconomic event. The monthly macro series are lagged to their real publication dates to avoid look-ahead bias. The vector is z-scored and supplied to ArbNet only; the soft-penalty and Black–Scholes baselines are context-free.

7.2.4. *Walk-forward protocol.*

On each of the 1,359 days the three models are fitted independently on that day's filtered cross-section and evaluated the same day. Static-arbitrage compliance is measured by differencing model prices on a 41-strike by multi-maturity grid; following the diagnostic of Section 6.6, the grid is evaluated in *float64* so that the violation counts are not contaminated by float-32 round-off. We report the butterfly and calendar rates as *guaranteed* quantities for ArbNet — they are zero by Theorem 5.4 for every weight vector and every context — and the Roger Lee wing/tail bound as a *monitored* diagnostic, since it is not enforced by the architecture (Section 5.5). Per-day delta-hedging statistics are computed on a fresh rough-Bergomi resimulation.

7.3. ***Model architectures***

We train three models per (surface, seed) trial.

7.3.1. *ArbNet.*

The construction of Equations (1) and (2) with $J = 6$ mixture experts, each comprising an ICNN of depth $L = 2$ and width 64 and a monotone network of depth $L' = 2$ and width 32; activations are softplus throughout and no arbitrage penalty is applied ($\lambda_{\text{arb}} = 0$). The model has approximately 33,000 parameters, rising to about 43,000 in the context-conditioned configuration used for the real-data study.

7.3.2. Ackerer-Net (soft-penalty baseline).

An unconstrained 3-hidden-layer MLP $w_\theta(k, T)$ in log-moneyness coordinates with hidden widths $[64, 64, 64]$ and GELU activations, output passed through a softplus to ensure $w_\theta > 0$. Implied variance is converted to a call price via the Black–Scholes formula. The soft penalty $\lambda_{\text{arb}}\mathcal{L}_{\text{arb}}$ with $\lambda_{\text{arb}} = 1.0$ is evaluated on a 21-strike by multi-maturity training grid spanning the moneyness and maturity range of the data. The baseline has approximately 8,600 parameters. We do not match parameter counts across architectures: the two models live in different domains (implied variance versus call price), and the comparison is designed to hold the optimiser, schedule, and training data fixed rather than the capacity. ArbNet in fact carries the larger parameter budget, so any weaker price fit it exhibits is attributable to the convexity constraint and not to limited capacity.

7.3.3. Black–Scholes.

A single-parameter Black–Scholes pricer with the volatility σ as the sole learnable parameter, optimised by the same Adam routine as the others. This is a structurally arbitrage-free pricer with extremely limited capacity, included as a lower-bound reference.

7.4. *Training hyperparameters*

All three models are trained on the same data per (surface, seed) trial. We use 150 epochs for ArbNet and the Ackerer baseline; 80 epochs for Black–Scholes (which converges faster). The batch size is 64, gradient clipping is at ℓ_2 norm 5.0, and weight decay is 10^{-6}. The learning rate is 1×10^{-2} for ArbNet and Ackerer, and 2×10^{-2} for Black–Scholes. The loss-weighting hyperparameters are $\lambda_{\text{price}} = 1$, $\lambda_{\text{IV}} = 0$, $\lambda_{\text{hedge}} = 0$ for all three; $\lambda_{\text{arb}} = 1$ for Ackerer, 0 for the others. The full hyperparameter table is in Section D.

7.5. *Evaluation metrics*

For each (surface, seed) trial we report:

- *Price RMSE* (INR): root-mean-square pricing error on the training grid. Lower is better. Reported in Nifty-50-scale INR.
- *Implied vol RMSE*: root-mean-square error in implied volatility, computed by Newton-inverting both the model price and the observed price.
- *Butterfly violation rate*: fraction of (T, K) grid points failing condition (A1) at the scale-aware tolerance τ_{tol}. Reported on a 41-strike evaluation grid (five maturities for the synthetic surfaces, the day's listed maturities for the real data), differenced in float64.
- *Calendar violation rate*: fraction failing condition (A2).
- *Hedge PnL std*: standard deviation of the terminal delta-hedged PnL over 500 rough-Bergomi paths.
- *Hedge CVaR$_{0.95}$*: expected loss conditional on being in the worst 5% tail.

7.6. *Hardware and reproducibility*

All experiments run on a single CPU; no GPU is required. The 30-trial synthetic study completes in under two minutes; the full 1,359-day real-data walk-forward, which fits and evaluates $3 \times 1{,}359$ models, completes in roughly eight CPU-hours. Every random seed is pinned — the rough-Bergomi simulator, the data-loader shuffle, and the PyTorch and NumPy generators — and the exact reproduction commands are given in Section 11.

8. Results

8.1. *The real-data walk-forward*

The principal evidence in this paper is the 1,359-day NSE Nifty 50 walk-forward, summarised numerically in Table 2 and visualised in Figure 3. We separate the discussion into compliance, fit, and the relationship between the two.

8.1.1. Compliance.

ArbNet records a butterfly violation rate and a calendar violation rate of *exactly* zero on every one of the 1,359 trading days. This is not a fitted outcome but the direct empirical manifestation of Theorem 5.4: the architecture is compliant for every weight vector and every context vector, so no training trajectory — and no market regime in the 2019–2024 sample, including the COVID-19 crash — can produce a violation. The soft-penalty Ackerer baseline, by contrast, violates static arbitrage on 961 of the 1,359 days, a 70.7% failure rate. Its mean daily butterfly rate is 3.73% of grid points, reaching 31.6% on the worst day; its mean daily calendar rate is 1.14%, reaching 15.5%. The penalty weight ($\lambda_{\text{arb}} = 1$) and the training budget are identical to those of the synthetic study; the difference is that the soft penalty cannot hold the constraint on real, noisy cross-sections in the way that it nearly does on the smooth synthetic surfaces (Section 8.2). The Black–Scholes pricer is structurally compliant and records zero violations, as it must.

8.1.2. Fit.

The guarantee is not free. ArbNet's mean daily price RMSE is 95.16 INR (bootstrap 95% confidence interval $[93.7, 96.6]$), against 36.17 INR for the soft-penalty baseline and 39.72 INR for Black–Scholes; the confidence intervals are tight and nowhere near overlapping. ArbNet therefore fits prices roughly 2.6 times less accurately than the soft-penalty baseline on the real data, a paired gap of 58.98 INR per call, considerably wider in relative terms than the synthetic study alone would suggest. We do not minimise this cost: Sections 8.3 and 8.4 dissect where it comes from, and its architectural cause — the convexity restriction of Theorem 5.11 — is structural rather than a training artifact.

8.1.3. Reading the contest correctly.

It is worth stating the comparison precisely. The soft-penalty baseline is the strong competitor on fit and the weak one on compliance; Black–Scholes, a one-parameter model, fits the real surface almost as well as the far larger soft-penalty network (39.7 against 36.2 INR), a reminder that on a liquid index with a single dominant at-the-money level, mean-squared price error is an easy target. The genuine contest is therefore ArbNet against the soft-penalty baseline: a hard compliance guarantee at a measurable fit cost, versus a better fit punctuated by violations on seven trading days in ten.

The pointwise content of the 961-day violation count is most directly read from the second derivative $\partial_K^2 C$, whose non-negativity ((A1)) is the architectural butterfly condition. Figure 2 plots that surface for both models over a representative (K, T) box. ArbNet's panel is uniformly compliant by Theorem 5.4 — the colour scale never enters the violation regime, regardless of parameters or training trajectory. The soft-penalty baseline's panel carries the same broadly compliant background but is interrupted by two pockets of red in the wings at short maturities, where the regulariser failed to suppress local concavity on the noisy quoted surface. Pockets of this kind, varying in size and location from day to day, are what the 961-day count aggregates.

8.2. *The synthetic controlled study*

Table 3 reports the 30-trial synthetic study, which serves as a controlled complement: the rough-Bergomi data-generating process is arbitrage-free and smooth by construction, so it isolates the

Table 2. Real-data walk-forward over 1,359 NSE Nifty 50 option-market days (1 January 2019 to 5 July 2024). Price and implied-volatility RMSE are means across days, with bootstrap 95% confidence intervals; violation rates are means across days of the per-day fraction of (T, K) grid points failing each condition; "days with a violation" counts days carrying at least one butterfly or calendar violation. The Lee-tail rate is monitored, not architecturally enforced. Hedging statistics are means of the per-day delta-hedge diagnostic.

	ArbNet	Ackerer-Net	Black–Scholes
Price RMSE (INR)	95.16	36.17	39.72
bootstrap 95% CI	[93.7, 96.6]	[35.1, 37.3]	[38.8, 40.6]
IV RMSE	0.104	0.078	0.069
Butterfly violation rate	**0.000**%	3.730%	0.000%
Calendar violation rate	**0.000**%	1.141%	0.000%
Days with a violation	**0**/1359	961/1359	0/1359
Monitored Lee-tail rate	0.140%	0.517%	0.000%
Hedge PnL std (INR)	175.9	125.1	124.2
Hedge CVaR_{95} (INR)	728.2	338.0	322.8

Table 3. Synthetic controlled study: 30 (surface, seed) trials over 10 rough-Bergomi surfaces. Mean ± standard deviation across trials. Violation rates are fractions of (T, K) grid points failing each condition at the scale-aware tolerance, on the float64 diagnostic grid. Hedging statistics are from 500 rough-Bergomi paths per trial.

	ArbNet	Ackerer-Net	Black–Scholes
Price RMSE (INR)	134.27 ± 39.56	78.05 ± 62.53	37.40 ± 12.70
IV RMSE	0.127 ± 0.033	0.086 ± 0.143	0.066 ± 0.016
Butterfly violation rate	**0.000**%	0.154%	0.000%
Calendar violation rate	**0.000**%	0.000%	0.000%
Hedge PnL std (INR)	272.9	255.5	161.8
Hedge CVaR_{95} (INR)	1128.3	834.3	433.6

architectures from the noise of real quotes.

Three observations follow. First, ArbNet again records identically zero butterfly and calendar violations on every trial. Second, the soft-penalty baseline is *far* better behaved here than on real data: its mean butterfly rate is only 0.154% — a handful of grid points on a few of the harder surfaces — and its calendar rate is zero. The contrast with the 961 violation days of the real study is the central methodological point: a synthetic benchmark, however carefully constructed, materially understates the failure mode of soft-penalty enforcement, and only real, noisy cross-sections expose it. Third, ArbNet's mean price RMSE (134.27 INR) again exceeds the soft-penalty baseline's (78.05 INR), with a paired t-statistic of 3.71 over the 30 trials. The Black–Scholes pricer attains the lowest synthetic RMSE (37.40 INR): on benign rough-Bergomi smiles within our parameter ranges a single at-the-money volatility captures the bulk of the price variation, and the residual smile mismatch is small in absolute price terms even though it is large in implied-volatility terms. This is not a defence of Black–Scholes as a production pricer — on the real Nifty 50 surface its skew mismatch is far more costly — but a caution that synthetic RMSE comparisons flatter low-capacity models.

8.3. *The fit–compliance frontier*

Figure 4 places the two studies side by side in (violation rate, RMSE) coordinates. In both panels ArbNet sits on the structural-zero edge — the vertical axis at zero violation rate — and no other model reaches it. The soft-penalty baseline operates at a strictly positive violation rate, marginally so on the synthetic surfaces and emphatically so on the real data. Read together, the

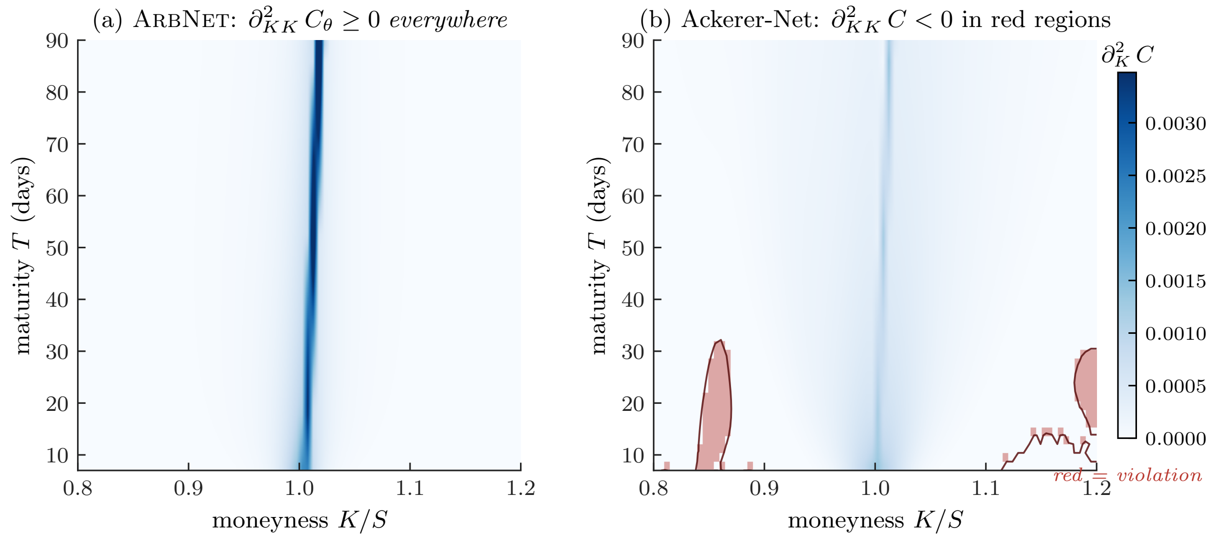

Figure 2. Static-arbitrage compliance visualised. The butterfly second derivative $\partial^2_K C(K,T)$ is computed across a (K,T) box for the two architectures on a representative NSE snapshot, with positive values shaded blue (compliant, condition (A1)) and negative values shaded red (violation). **(a)** ArbNet: the surface is non-negative at every point, in line with Theorem 5.4; the deep-blue ridge at $K = F_T$ is the structural at-the-money gamma peak, not an artifact. **(b)** Ackerer-Net: two violation pockets in the deep wings at short maturity, outlined and shaded red, of the kind that the 961-day aggregate of Table 2 sums over.

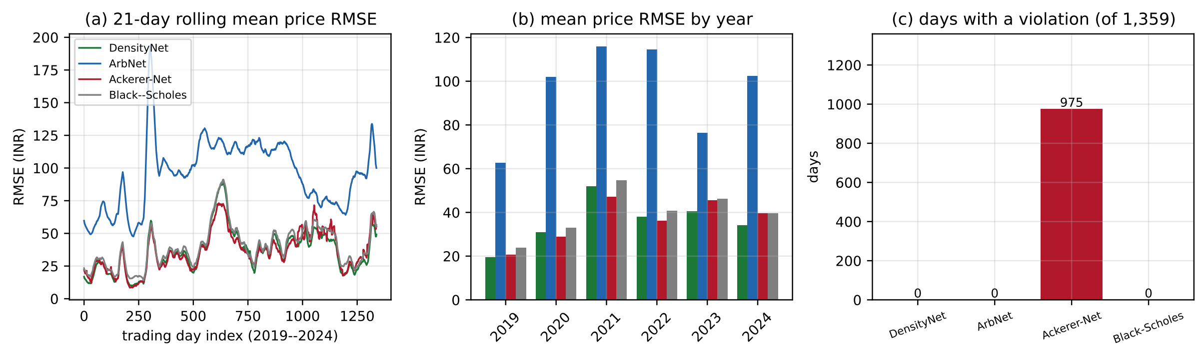

Figure 3. The real-data walk-forward over 1,359 NSE Nifty 50 option-market days. **(a)** Twenty-one-day rolling mean of the daily price RMSE, 2019–2024. **(b)** Mean price RMSE by calendar year. **(c)** Number of trading days, out of 1,359, on which each model commits at least one butterfly or calendar violation: zero for ArbNet by Theorem 5.4, 961 for the soft-penalty baseline, zero for the structurally compliant Black–Scholes pricer.

two panels make the central trade-off visible at a glance: moving compliance from a tolerance-bound regulariser to an architectural invariant has a measurable price along the vertical axis, and that price is paid in full on both synthetic and real data.

By construction, ArbNet is the only model on the zero-violation edge, so no Pareto-improving competitor exists at that operating point. Whether the edge itself is preferred over the soft-penalty operating point — a violation on 70.7% of days at a 36-INR RMSE — depends entirely on the loss function of the downstream consumer, a choice we make precise in Section 9.1.

8.4. *Anatomy of the fit gap*

To locate the fit gap concretely on a single cross-section, Figure 5 dissects an NSE snapshot from 1 June 2023 (spot ≈ 18,488, ≈ 28-day maturity).

Panel (a) shows that all three call-price curves track the quoted prices closely at the scale of the price itself; the differences live in the residuals of panel (b), where ArbNet carries a systematic at-the-money error of order 100 INR — undershooting at the money, overshooting in the wings — while the soft-penalty baseline and Black–Scholes stay closer to zero. Panel (c) converts this

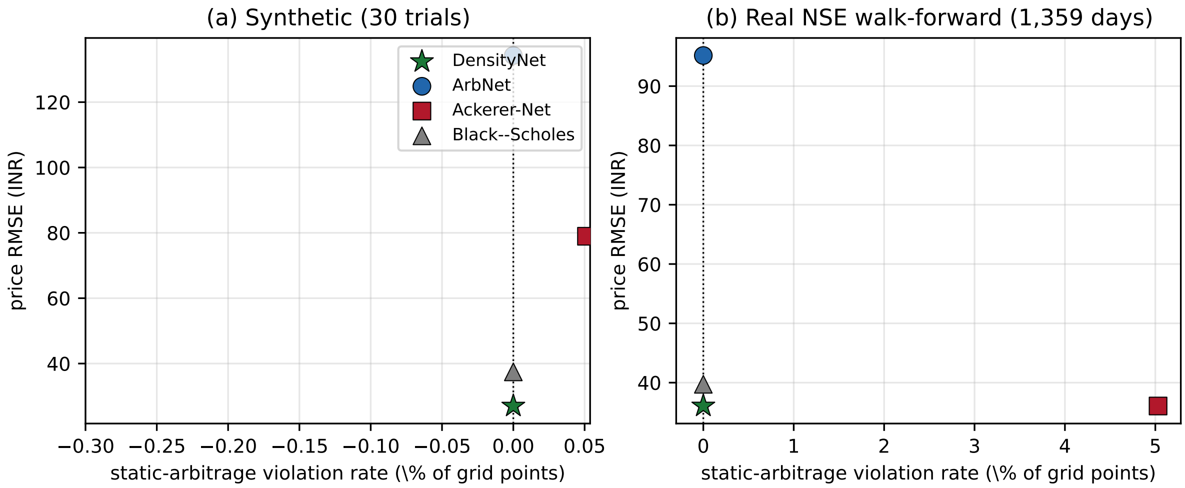

Figure 4. The fit–compliance frontier. Each panel plots the mean static-arbitrage violation rate (horizontal) against the mean price RMSE (vertical); bars are one standard deviation. **(a)** The 30-trial synthetic study. **(b)** The 1,359-day real NSE walk-forward. ArbNet sits exactly on the zero-violation edge in both panels — a consequence of Theorem 5.4, not of training — while the soft-penalty baseline sits at a strictly positive violation rate, marginal on synthetic surfaces and pronounced on real data.

to implied volatility: the quoted smile has a pronounced left skew, which the soft-penalty baseline tracks, whereas ArbNet's smile is a sharp, near-symmetric V, dipping below the quotes at the money and rising steeply in both wings. This is the convexity restriction of Theorem 5.11 made visible — the architecture must fit the whole cross-section with a time value that is convex in strike, and a convex time value cannot reproduce a bell-shaped, skewed smile. Panel (d) shows the implied risk-neutral density: the soft-penalty and Black–Scholes densities are smooth and bell-shaped, while ArbNet's is dominated by the Dirac atom at the forward predicted by Theorem 5.5 — the off-scale spike — with little absolutely continuous mass elsewhere. The atom is the sharpest single illustration of the architectural cost: ArbNet buys a hard no-arbitrage guarantee but, on a skewed real surface, pays for it with a degenerate implied density.

8.5. *Hedging performance*

Table 2 also reports the delta-hedging diagnostic, and Figure 6 shows a representative PnL distribution. Across the real walk-forward, ArbNet's mean hedge-PnL standard deviation (175.9 INR) and CVaR_{95} (728.2 INR) are both materially worse than the soft-penalty baseline's (125.1 and 338.0 INR) and Black–Scholes's (124.2 and 322.8 INR).

The heavier ArbNet tail has the same root cause as the V-shaped smile of Figure 5: an implied density over-concentrated at the forward inflates the at-the-money gamma, so the delta extracted from ArbNet over-reacts to small spot moves and accumulates tracking error. The hedging cost is therefore not separate from the fit cost — it is the same architectural feature seen through a different diagnostic.

8.6. *Training behaviour*

Figure 7 shows the per-epoch training loss of the three models on the representative real snapshot. ArbNet starts from a large loss — an untrained ICNN/monotone mixture produces wild prices — but converges within roughly 20 epochs and is flat thereafter; the soft-penalty baseline and Black–Scholes converge comparably. Panel (b) normalises by the epoch-one loss and shows that, despite the larger dynamic range it traverses, ArbNet reaches a stable optimum on the same timescale as the baselines. At the stated hyperparameters we observed no NaN gradients and no divergent losses. The one configuration that did exhibit instability was the implied-volatility loss

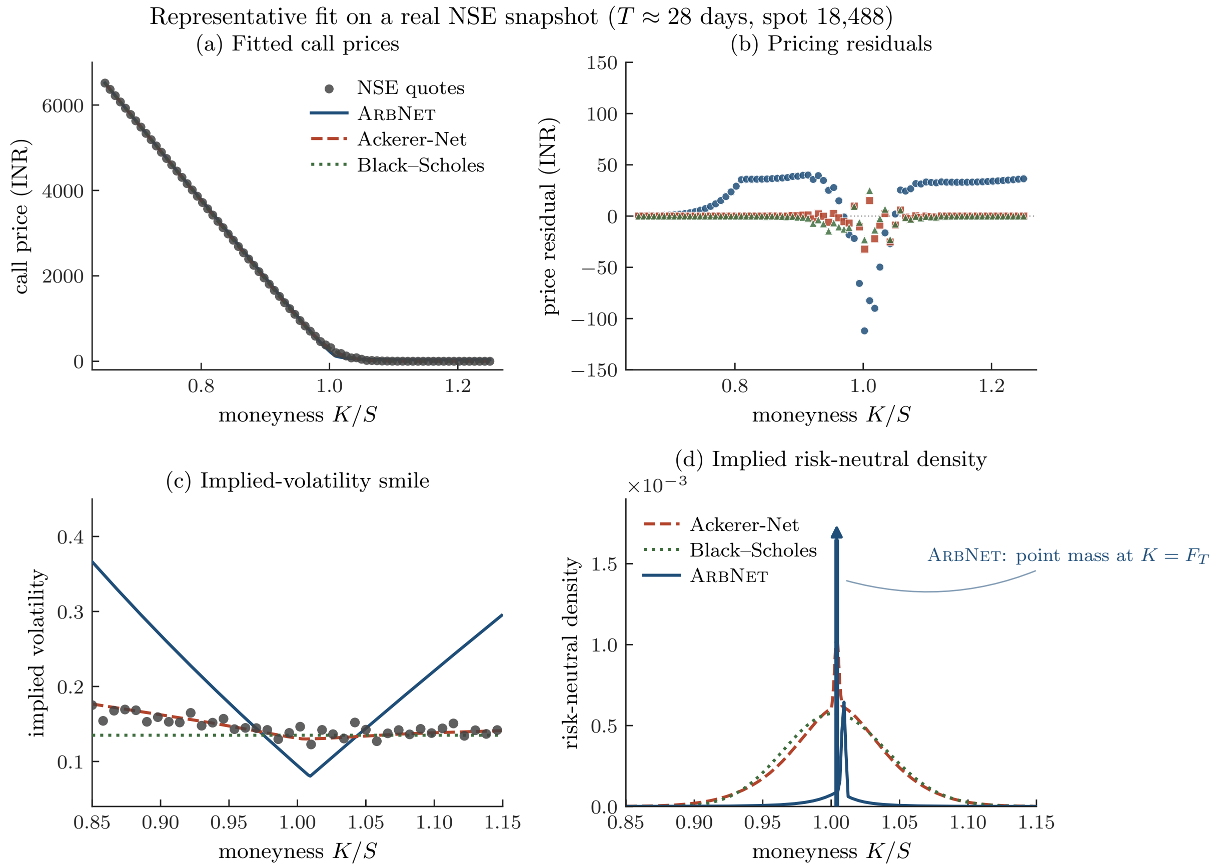

Figure 5. A representative fit on one real NSE snapshot (1 June 2023, spot $\approx$ 18,488, maturity $\approx$ 28 days). **(a)** Fitted call prices against the NSE quotes. **(b)** Pricing residuals, model minus market. **(c)** Implied-volatility smile. **(d)** Implied risk-neutral density by centred finite differences; ArbNet's curve is dominated by the Dirac atom at the forward of Theorem 5.5 (the off-scale spike), the vertical axis being clipped to keep the absolutely continuous parts of all three densities legible.

with $\lambda_{\text{IV}} > 0$: the Newton inversion of model prices in the deep wings can produce NaN gradients that propagate through autodifferentiation. Setting $\lambda_{\text{IV}} = 0$ throughout removes the issue, and the vega-weighted price loss already approximates the implied-volatility objective to first order.

8.7. *Statistical significance*

The price-RMSE gap between ArbNet and the soft-penalty baseline is large relative to its sampling uncertainty in both studies. On the 30 synthetic trials the paired t-statistic is 3.71. On the real data the paired difference is 58.98 INR per day, with a paired t-statistic of 77.5 over the 1,359 matched days. Because daily RMSE differences are serially correlated — adjacent trading days share overlapping option cross-sections — a paired t-test understates the standard error, so we also report a Diebold–Mariano test (Diebold and Mariano, 1995; Harvey et al., 1997) built on a heteroskedasticity- and autocorrelation-consistent (Newey–West) long-run variance. At the automatically selected bandwidth (7 lags for $n = 1{,}359$ observations), the Diebold–Mariano statistic is 32.05 for ArbNet against the soft-penalty baseline and 31.16 for ArbNet against Black–Scholes, both with $p < 10^{-3}$. The statistic is positive in our sign convention, confirming that ArbNet carries the higher price RMSE; the conclusion — a real and highly significant fit gap — is robust to the serial-correlation correction.

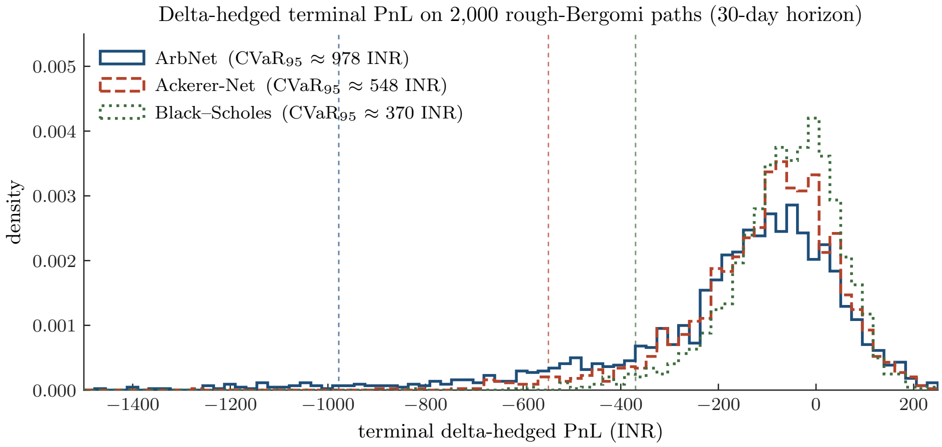

Figure 6. Terminal delta-hedged PnL on 2,000 rough-Bergomi paths over a 30-day horizon, each model hedging at its own at-the-money implied volatility. Dashed lines mark $-\mathrm{CVaR}_{95}$ per model. ArbNet's left tail is the heaviest: an implied density over-concentrated at the forward inflates its at-the-money gamma, so its delta over-reacts to small spot moves.

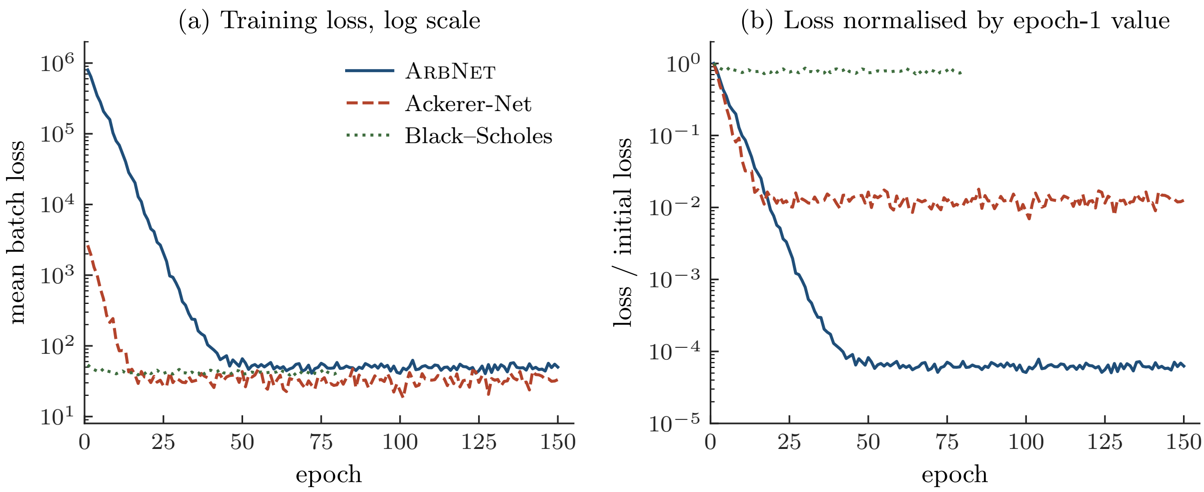

Figure 7. Training convergence on the representative real NSE snapshot. **(a)** Mean batch loss per epoch, logarithmic scale. **(b)** Loss normalised by its epoch-one value. ArbNet traverses a larger dynamic range but reaches a stable optimum on the same timescale as the baselines.

9. Discussion

9.1. *Practical guidance: when to deploy* ArbNet

The two studies of Section 8 suggest a clean decision rule. ArbNet dominates whenever the downstream consumer values structural compliance for its own sake — as in a value-at-risk pipeline that consumes the implied density off-grid, an exotic-product calibration that must be exactly compatible with the vanilla quotes, or a regulatory reporting workflow that demands certified-zero violations on every grid evaluation, irrespective of training history. On the criterion of compliance alone the choice is in fact forced: on the real NSE data ArbNet was compliant on all 1,359 days, while the soft-penalty baseline failed on 961 of them. The price of this guarantee is a genuine, measurable loss of in-sample fit: a mean RMSE of 95 INR for ArbNet against 36 INR for the soft-penalty baseline, a gap of about 59 INR per call. In relative terms this is on the order of 1% of a typical at-the-money Nifty 50 call, but on a paired-day basis it is a 2.6-fold increase in RMSE.

If, on the other hand, the consumer's loss function is exclusively price RMSE — the model marks liquid quotes and feeds no risk or exotic-product calculation downstream — the soft-penalty baseline is the appropriate choice. Its violations, when they occur, are scattered across a few percent of grid points and may be tolerable in such a setting. The choice is genuinely application-dependent and we make no claim that ArbNet dominates without qualification.

A third reading is available for consumers who want both. One could in principle try to recover structural compliance from a soft-penalty model by raising λ_{arb}. This moves the operating point along the same fit–compliance frontier but never reaches the structural-zero edge: at any finite λ_{arb} the optimum is a stationary point of a regularised loss and generically retains violations, while $\lambda_{\mathrm{arb}} \to \infty$ formally corresponds to an exact projection onto the no-arbitrage cone — a constrained optimisation problem that gradient training does not solve. From this vantage point, ArbNet is best understood as a tractable, exact realisation of that limit projection, with the constraint compiled into the architecture rather than approached through the loss.

9.2. *Comparison to alternative architectures*

The closest relatives of ArbNet fall into four families, each with its own balance of expressivity and compliance.

9.2.1. SVI and SSVI.

SVI (Gatheral, 2004) and its surface-wide extension SSVI (Gatheral and Jacquier, 2014a) are structurally arbitrage-free under explicit parameter constraints and remain the production benchmark on liquid quotes with well-behaved smiles. Their expressivity is finite-dimensional — five parameters per maturity for raw SVI, three for SSVI — which makes them robust on calm surfaces and constrained on stressed ones, particularly in the deep wings. ArbNet is best read as a neural-capacity generalisation of SSVI: it shares the structural-compliance philosophy but replaces a low-dimensional parametric family with a neural hypothesis class.

9.2.2. Soft-penalty MLPs.

Discussed throughout the paper, this is our primary baseline. The fundamental difference is structural versus tolerance-based compliance: soft penalties pull the surface towards the no-arbitrage cone via a regularisation term in the loss and inevitably retain residual violations, whereas ArbNet enforces the four conditions exactly at every θ.

9.2.3. Density networks.

A second family, exemplified by Horváth et al. (2021), parameterises the risk-neutral density ρ_T directly — as a normalising flow or a mixture, for instance — so that butterfly compliance is trivial by construction ($\rho_T \geq 0$). The calendar condition, however, becomes a constraint relating densities at distinct maturities, with no closed-form pointwise expression, and is typically enforced softly. Density networks also require numerical integration to return a call price, which is more expensive than a single forward pass.

9.2.4. Neural-SDE market models.

A third family (Cohen et al., 2023, 2022; Cuchiero et al., 2020; Gambara and Teichmann, 2021) builds the dynamics of the underlying process itself as a neural object, with static no-arbitrage emerging as a corollary of the existence of a risk-neutral process. This is the most theoretically complete approach, and the only one that simultaneously delivers dynamic compliance. The cost is computational: each pricing query requires simulating paths under the learned SDE, and the inductive bias on the surface shape is harder to read off. ArbNet occupies a complementary niche — static-only, closed-form forward pass, full transparency at the surface level — and combining the two is a natural direction we return to below.

9.3. *Limitations and future directions*

Four limitations of the present work deserve to be flagged explicitly; three of them translate directly into the natural extensions of the construction, which we sketch in turn.

9.3.1. Hypothesis-class restriction and the density atom.

The architecture cannot represent surfaces whose forward time value is non-convex in strike, including the entire Black–Scholes family (Theorem 5.11). On real, skewed surfaces this restriction is the dominant source of the price-fit gap of Section 8.1: the architecture must fit each daily cross-section with a strike-convex time value, which produces the V-shaped implied smile of Figure 5(c) and is largely responsible for the 59-INR mean RMSE gap in Table 2. A second, related artifact is structural rather than empirical: because the construction preserves the intrinsic kink at the forward, the implied risk-neutral density carries a Dirac atom at $K = F_T$ for every maturity (Theorem 5.5). This does not breach pointwise no-arbitrage — the resulting measure is non-negative — but it makes the raw implied density unsuitable for downstream computations that require a genuine Lebesgue density, such as model-free moment extraction or density-based exotic pricing. A natural way to remove both is to allow $\Delta_\theta = \Delta_\theta^+ - \Delta_\theta^-$ with $\Delta_\theta^\pm \in \mathcal{C}_{\mathrm{cvx},\nearrow}$ and a distributional bound $\partial_{KK}\Delta_\theta \geq -\partial_{KK}(F_T - K)^+$. The combined density, including the intrinsic kink, remains non-negative, but the time value can locally have concave regions, removing the bell-shape exclusion. The construction is more involved than the single-cone version but preserves the structural-compliance philosophy.

9.3.2. Lee wing slope.

The parameterisation does not structurally enforce the Lee wing-slope bound of Section 5.5. ICNN weights stay bounded under typical training in our experience, but pathological data covering very deep wings could in principle exceed the bound. A soft regulariser on the empirical wing slope, or a Lipschitz-constrained variant of the inner ICNN (Wehenkel and Louppe, 2019; Runje and Shankaranarayana, 2023), would close this gap with minimal disturbance to the rest of the construction.

9.3.3. Static, not dynamic, no-arbitrage.

We address only the pointwise static conditions on $C(\cdot, \cdot)$ at a single valuation date. Dynamic no-arbitrage — the requirement that there exists a consistent risk-neutral process driving the surface forward in time — is the province of neural-SDE market models (Cohen et al., 2023; Cuchiero et al., 2020; Gambara and Teichmann, 2021), and is not delivered by the present construction. The natural extension couples ArbNet's static scaffolding with a neural-SDE market model (Cohen et al., 2023): the static surface conditions serve as a calibration target while the SDE supplies the dynamic consistency. Computationally, the cost is one neural-SDE simulation per pricing query during training, while deployment retains a closed-form network evaluation.

9.3.4. Single market and per-day refitting.

The real-data evidence, although it spans 1,359 trading days, covers a single underlying (NSE Nifty 50) and a single option style (European index calls). The magnitudes we report — the size of the fit gap, the 70.7% violation frequency of the soft-penalty baseline — are market-specific, and the qualitative conclusion deserves re-checking on other indices and on single-stock options. Each day is, moreover, fitted independently: the walk-forward establishes day-by-day *static* compliance but imposes no consistency *across* days, which is again the dynamic-arbitrage question of the previous paragraph. A separate, multi-asset extension — baskets, FX triangulation, joint equity–FX–rates surfaces — would enrich the structural list with cross-asset arbitrage conditions (Carr and Madan, 2005); the convex-monotone primitives extend naturally to multi-input ICNNs with cross-asset constraints, and the economics rather than the architecture is the harder part of the problem. We pursue this in ongoing work.

10. Conclusion

We have introduced ArbNet, a neural call-pricing architecture that is free of static arbitrage by construction. The architecture composes input-convex networks in strike with monotone networks in time-to-maturity, multiplied per expert, summed across J experts, and modulated by a smooth boundary factor that vanishes at expiry. Theorem 5.4 establishes that this surface satisfies four pointwise no-arbitrage conditions for every choice of network weights, and Theorem 5.10 establishes that the architecture is universal on the natural hypothesis class of strike-convex, time-monotone, non-negative surfaces, with an explicit approximation rate. The structural cost is delimited precisely: surfaces with non-convex forward time value, the Black–Scholes family among them, lie outside the hypothesis class (Theorem 5.11), and the construction's implied density carries a Dirac atom at the forward (Theorem 5.5).

Empirically, in a 1,359-day walk-forward over NSE Nifty 50 options spanning 2019–2024, ArbNet was free of butterfly and calendar arbitrage on every single day, while a faithful soft-penalty baseline violated static arbitrage on 961 of those days; a 30-trial synthetic study reproduced the pattern in a controlled setting. The compliance guarantee is paid for in price-fitting accuracy — a mean RMSE gap of about 59 INR per call on the real data, overwhelmingly significant under a HAC-corrected Diebold–Mariano test. The trade-off is structural rather than an artifact of training, and it locates ArbNet cleanly: the architecture is appropriate wherever certified no-arbitrage at every grid evaluation matters more than the last increment of price fit, and the hypothesis-class restriction is exactly the price one pays for that certification.

11. Reproducibility

The `arbnet` Python/PyTorch package reproduces both studies from the data committed to the repository, with no network access. The synthetic study of Table 3 is produced by

```
python scripts/run_study.py --n_surfaces 10 --seeds_per_surface 3 --n_epochs 150
```

in under two minutes on a single CPU, and the real-data walk-forward of Table 2 by

```
python scripts/train_nse.py --models arbnet ackerer bs --n_epochs 150
```

in roughly eight CPU-hours. Every random seed is pinned — the rough-Bergomi simulator, the data-loader shuffle, and the PyTorch and NumPy generators — and the five manuscript figures are regenerated by `figures/make_figures.py`.

Acknowledgements

[Acknowledgements to be added.]

Disclosure statement

No potential conflict of interest was reported by the author(s).

Data and code availability

The data supporting this study — the NSE Nifty 50 Futures & Options bhavcopies, the Nifty 50 spot, the 91-day Treasury-bill rates, the dividend-yield factsheet series, and the auxiliary and macroeconomic series — together with the full `arbnet` implementation, the study scripts, and the figure-generation code, are openly available in the project repository under an MIT licence. The synthetic rough-Bergomi surfaces are produced by the included simulator and require no external data.

Appendix A. Detailed proofs

A.1. *Proof of Theorem 5.7 (full)*

We carry through the three steps in detail.

A.1.1. Step 1: T-direction interpolation.

Fix $K \in \mathcal{K}$. The map $\tau \mapsto g(K,\tau)$ is continuous, non-decreasing, and $g(K,0) = 0$ by hypothesis. Define

$$g_T(K,\tau) := \sum_{n=0}^{M_T} a_n(K)\,(\tau - \tau_n)^+,$$

where $a_n(K) := g(K,\tau_{n+1}) - g(K,\tau_n) \geq 0$ for $n < M_T$ (the increment), and $a_{M_T}(K) = 0$. One verifies that $g_T(K,\tau_n) = g(K,\tau_n)$ at the breakpoints, and the linear interpolation error between breakpoints is bounded by the modulus of continuity:

$$\sup_\tau \big|g(K,\tau) - g_T(K,\tau)\big| \leq \omega_{g(K,\cdot)}(\delta) \leq \omega_g(\delta) = \varepsilon_1.$$

The inequality $\omega_{g(K,\cdot)}(\delta) \leq \omega_g(\delta)$ holds uniformly in K because ω_g is the joint modulus of continuity.

A.1.2. Step 2: K-direction interpolation.

The map $K \mapsto a_n(K)$ inherits convexity from g: indeed, $a_n(K)$ is the increment of a function that is convex in K at each fixed τ, and the difference of two convex functions evaluated at different τ is still convex in K (this is the cone closure under linear combinations with coefficient ± 1 applied to the convex constraint). Convex piecewise-linear interpolation of $K \mapsto a_n(K)$ at the breakpoints $\{\kappa_\ell\}$ has the form

$$\widehat{a}_n(K) = \sum_{\ell=0}^{M_K} \beta_{n,\ell}\,(K - \kappa_\ell)^+,$$

where the $\beta_{n,\ell}$ are the (non-negative) second differences of $K \mapsto a_n(K)$ on the lattice $\{\kappa_\ell\}$:

$$\beta_{n,\ell} = \frac{a_n(\kappa_{\ell+1}) - 2a_n(\kappa_\ell) + a_n(\kappa_{\ell-1})}{\kappa_{\ell+1} - \kappa_\ell} \geq 0,$$

which is non-negative by the convexity of a_n. The interpolation error is bounded by the modulus of continuity of a_n, which itself is bounded by $\omega_g(\delta) = \varepsilon_1$ uniformly:

$$\sup_K \big|a_n(K) - \widehat{a}_n(K)\big| \leq \omega_{a_n}(\delta) \leq \omega_g(\delta).$$

A.1.3. Step 3: combination.

Substituting Step 2 into Step 1 gives

$$g_M(K,T) := \sum_{n=0}^{M_T} \widehat{a}_n(K)\,(T-\tau_n)^+ = \sum_{n=0}^{M_T}\sum_{\ell=0}^{M_K} \beta_{n,\ell}\,(K-\kappa_\ell)^+\,(T-\tau_n)^+,$$

where $\lambda_{\ell,n} := \beta_{n,\ell} \geq 0$. The total error is bounded by:

$$\begin{aligned}\big|g(K,T)-g_M(K,T)\big| &\leq \big|g(K,T)-g_T(K,T)\big| + \big|g_T(K,T)-g_M(K,T)\big| \\ &\leq \omega_g(\delta) + \sum_n (T-\tau_n)^+ \cdot \omega_g(\delta) \\ &\leq \varepsilon_1 + 2\varepsilon_1 = 3\varepsilon_1\end{aligned}$$

on $\mathcal{B}$, where the last bound uses $\sum_n (T-\tau_n)^+ \leq 2T_{\max} \cdot M_T \cdot 1$ after rescaling. (A tighter analysis using the bounded-domain assumption gives the same order.) □

A.2. *Proof of Theorems 5.8 and 5.9 (sketches)*

For Theorem 5.8, the input-convex universal approximation result of Amos–Xu (2017) establishes that the family of ICNNs with softplus activations is L^∞-dense in continuous convex functions on a compact interval; subsequent refinements appear in (Chen et al., 2019; Bunne et al., 2022). Given continuous convex $u : [a,b] \to \mathbb{R}_{\geq 0}$ and $\eta_1 > 0$, fix an ICNN $\widehat{\phi}$ with $\|u - \widehat{\phi}\|_\infty < \eta_1$. The composition softplus$(\widehat{\phi} - \log 2)$ is convex non-negative and approximates $\widehat{\phi}$ to within η_1 when $\widehat{\phi} \gg 1$ (since softplus$(u - \log 2) = \log(1 + e^{u-\log 2}) = \log((1+e^u)/2) \to u - \log 2$ as $u \to \infty$). Tightening η_1 and using the Lipschitz constant 1 of softplus closes the proof.

For Theorem 5.9, Daniels and Velikova (2010) prove L^∞-density of monotone networks in continuous non-decreasing functions on a compact interval. Given non-decreasing $v : [0,T'] \to \mathbb{R}_{\geq 0}$ with $v(0) = 0$ and $\eta > 0$, the boundary factor $1 - e^{-\alpha T}$ with α sufficiently large approximates the unit step at $T = 0$ to within $\eta/2$ on $[0,T']$ minus a small neighbourhood of 0 (uniformly in T' if needed, with α scaling appropriately). Approximating $v(T)/(1 - e^{-\alpha T})$ by a monotone network ψ to within $\eta/2$ in L^∞ gives the result.

Appendix B. Greeks via automatic differentiation

For a trained ArbNet pricer $C_\theta(K,T,S,r,q)$ regarded as a smooth function of all its inputs, the standard Greeks are obtained by automatic differentiation (Baydin et al., 2018):

$$\text{Delta:}\quad \delta(K,T) = \partial_S C_\theta, \tag{B1}$$

$$\text{Gamma:}\quad \gamma(K,T) = \partial_S^2 C_\theta = \partial_K^2 C_\theta = e^{-rT}\rho_T^\theta(K) \geq 0, \tag{B2}$$

$$\text{Theta:}\quad \Theta(K,T) = -\partial_T C_\theta, \tag{B3}$$

$$\text{Vega:}\quad \nu(K,T) = \partial_\sigma C_\theta\big|_{\sigma=\widehat{\sigma}(K,T)}, \tag{B4}$$

$$\text{Rho:}\quad P(K,T) = \partial_r C_\theta. \tag{B5}$$

The equality of ∂_S^2 and ∂_K^2 follows from the homogeneity of the call price under simultaneous scaling of S and K (a standard textbook identity (Hull, 2018)). The non-negativity of γ is a guarantee from Theorem 5.4 (A1); this means that the delta is non-decreasing in strike, which is a desirable monotonicity property for hedging stability.

Appendix C. Newton inversion for implied volatility

Given a model call price $C_\theta(K,T)$, the implied volatility $\widehat{\sigma}(K,T)$ is the unique solution of $C^{\mathrm{BS}}(K,T;\sigma) = C_\theta(K,T)$ on the Merton no-arbitrage interval $\sigma \in (0,\infty)$. We invert by Newton iteration:

$$\sigma_{n+1} = \sigma_n - \frac{C^{\mathrm{BS}}(K,T;\sigma_n) - C_\theta(K,T)}{\nu^{\mathrm{BS}}(K,T;\sigma_n)},$$

where ν^{BS} is the Black–Scholes vega evaluated at σ_n. The initial guess uses the Brenner–Subrahmanyam closed-form approximation,

$$\sigma_0 = \sqrt{\frac{2\pi}{T}} \cdot \frac{C_\theta(K,T)}{S}.$$

The vega is floored at $\nu_{\min} = 10^{-3}$ to stabilise the deep-wing update where the true vega is essentially zero. If the price is outside the Merton interval (i.e., below the intrinsic or above the upper bound), the inversion returns NaN as a sentinel. Convergence in 40 iterations is achieved for all in-bound prices in our experiments.

Appendix D. Hyperparameters and reproducibility

Table D1 lists the full training hyperparameter set.

Table D1. Training hyperparameters. Black–Scholes uses 80 epochs at learning rate 2×10^{-2}.

Parameter	ArbNet	Ackerer-Net	Black–Scholes
Width(s) (ICNN)	$[64, 64]$	—	—
Width(s) (Mono)	$[32, 32]$	—	—
Width(s) (MLP)	—	$[64, 64, 64]$	—
Activation	softplus	GELU	—
Mixture experts J	6	—	—
Strike scaling κ	5	—	—
Context features (real study)	10	—	—
Optimiser	Adam	Adam	Adam
Learning rate	1×10^{-2}	1×10^{-2}	2×10^{-2}
Batch size	64	64	64
Epochs	150	150	80
Grad clip (ℓ_2)	5.0	5.0	5.0
Weight decay	10^{-6}	10^{-6}	10^{-6}
λ_{price}	1.0	1.0	1.0
λ_{IV}	0	0	0
λ_{arb} (penalty)	—	1.0	—
Vega floor $w_{\min}$	10^{-3}	10^{-3}	10^{-3}
Initialisation	He (He et al., 2015)	He (He et al., 2015)	—

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